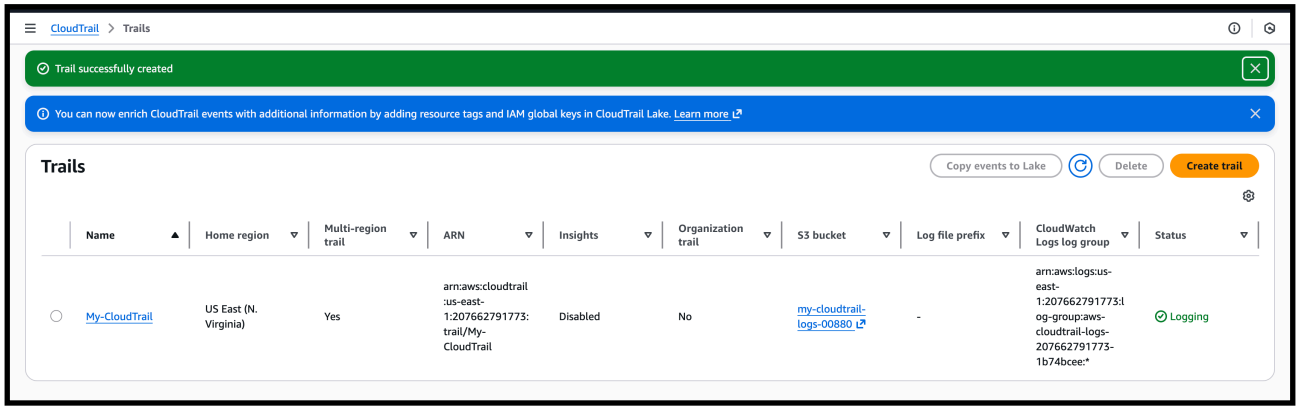
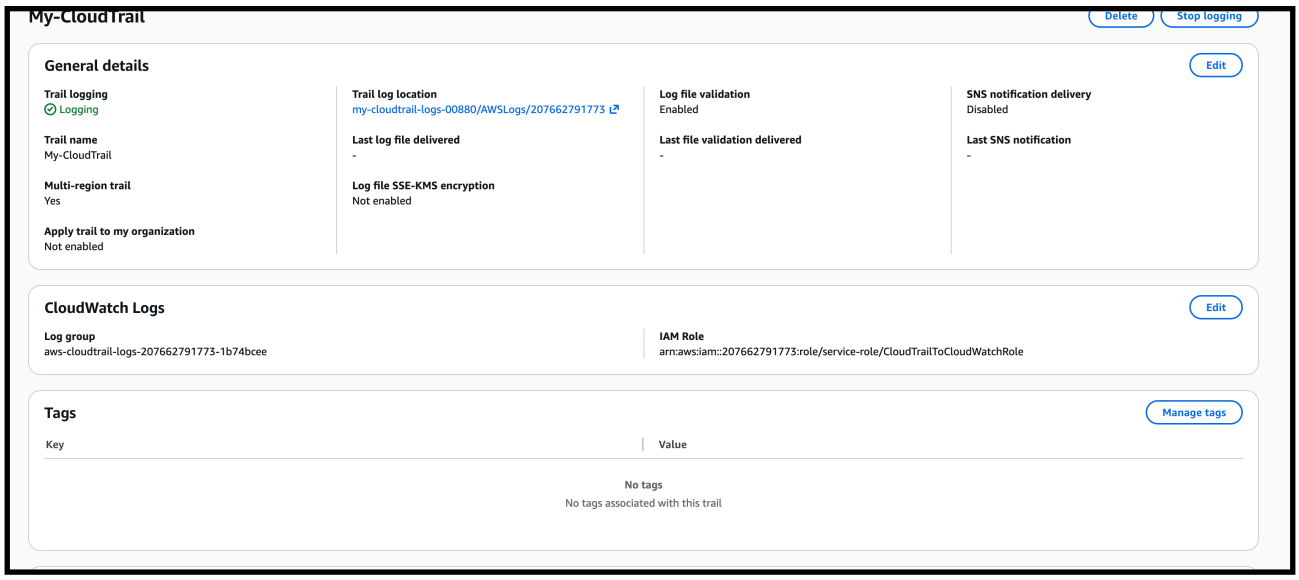
task on monitoring cloudtail cloud watch

1)Enable cloudtrail monitoring and store the events in s3 and cloudwatch log events.

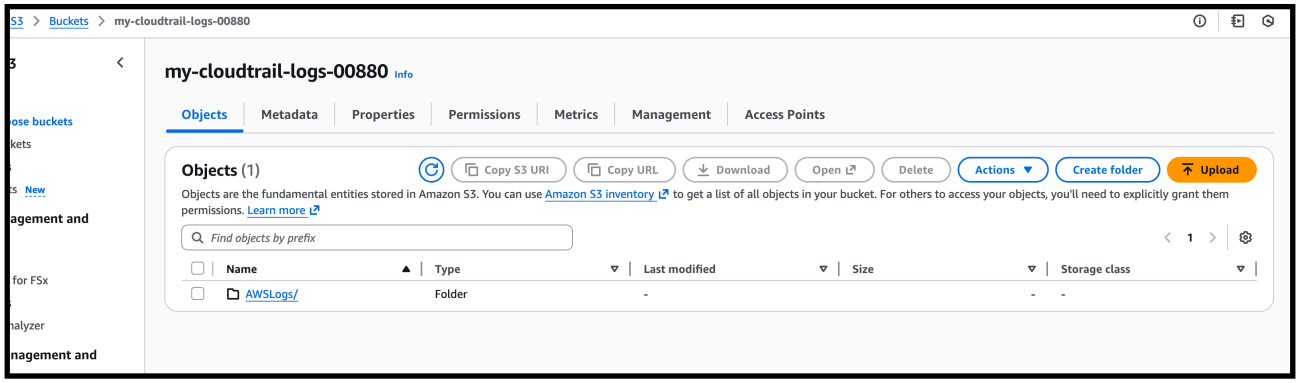
—> created a cloudtrail monitoring and storing the events in s3 bucket



—> created cloud trail details with s3 bucket detail and cloud watch details

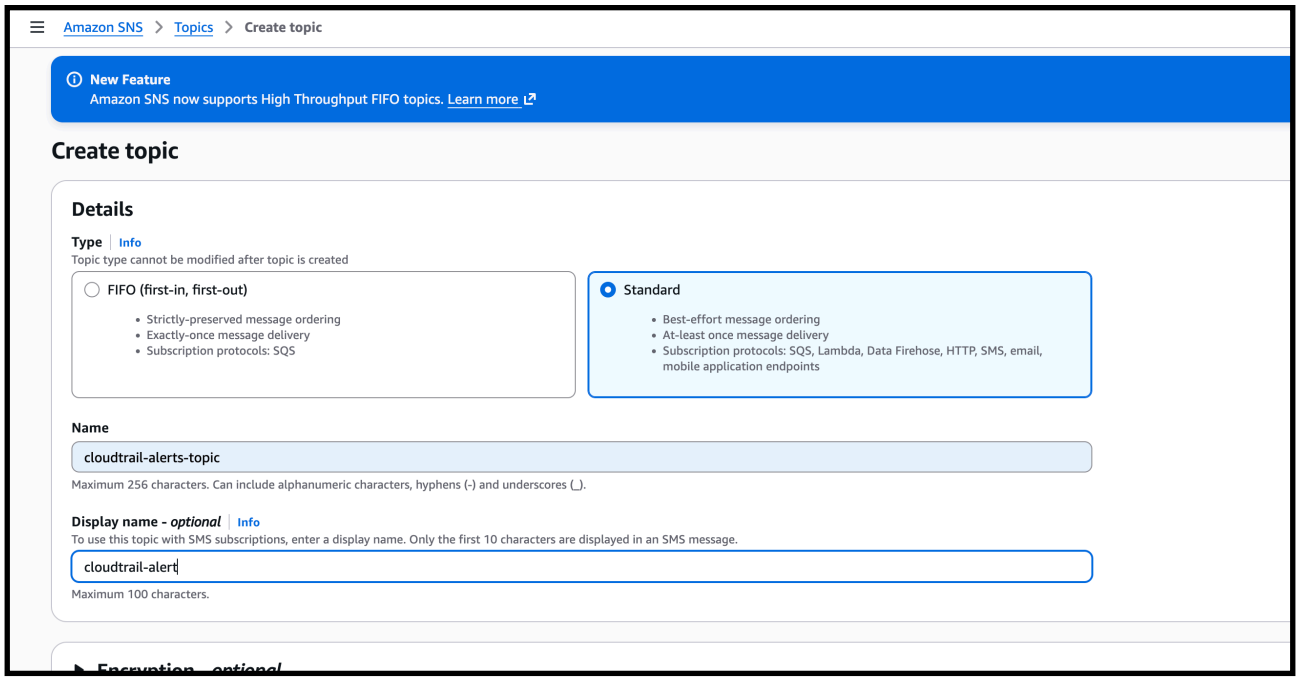


—> the cloudtrail logs are getting stored in s3 bucket

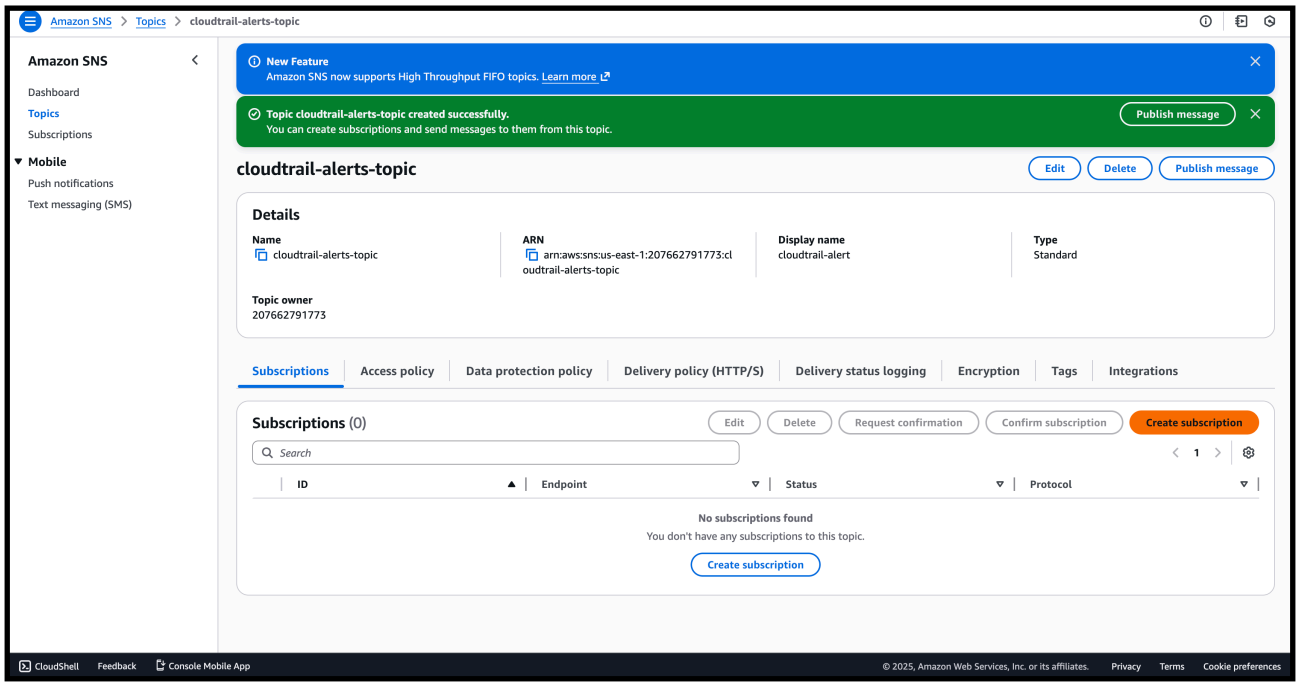


2)Enable SNS for cloudtrial to send alert on email.

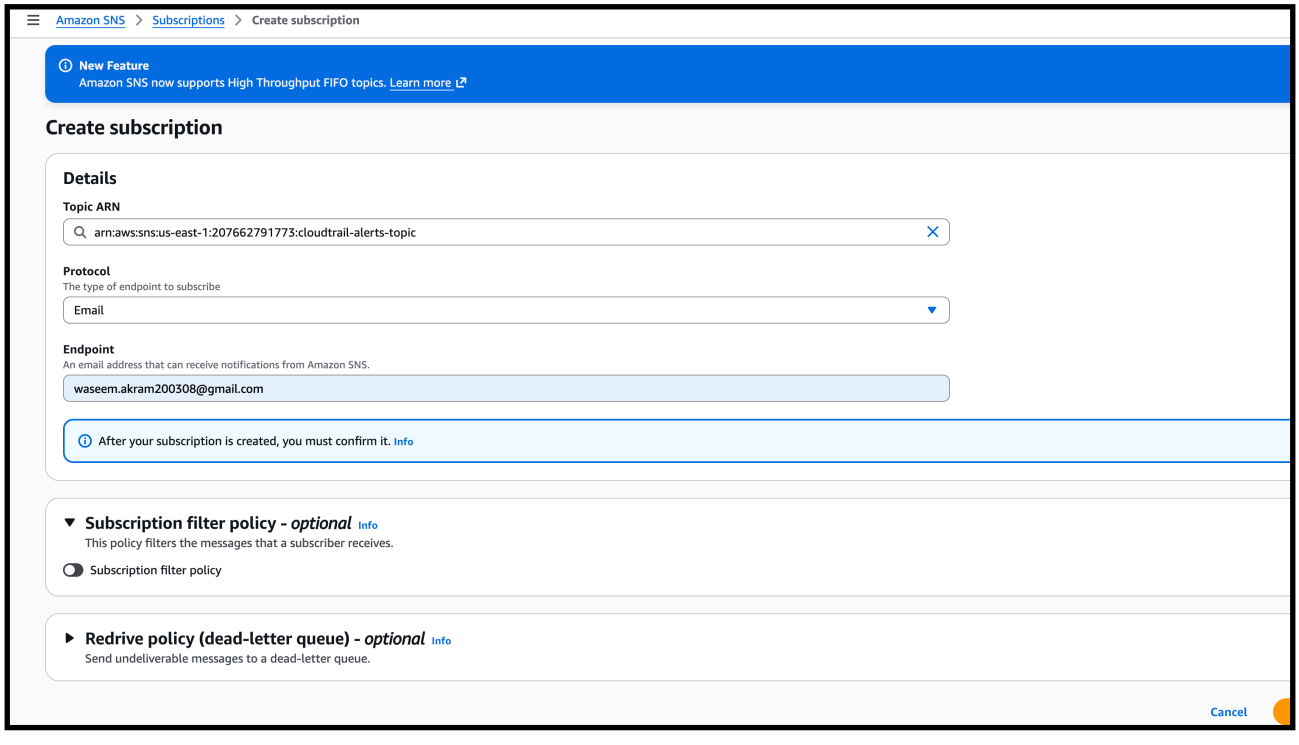
—> create a sns topic
—> give it a name cloudtrail-alerts-log



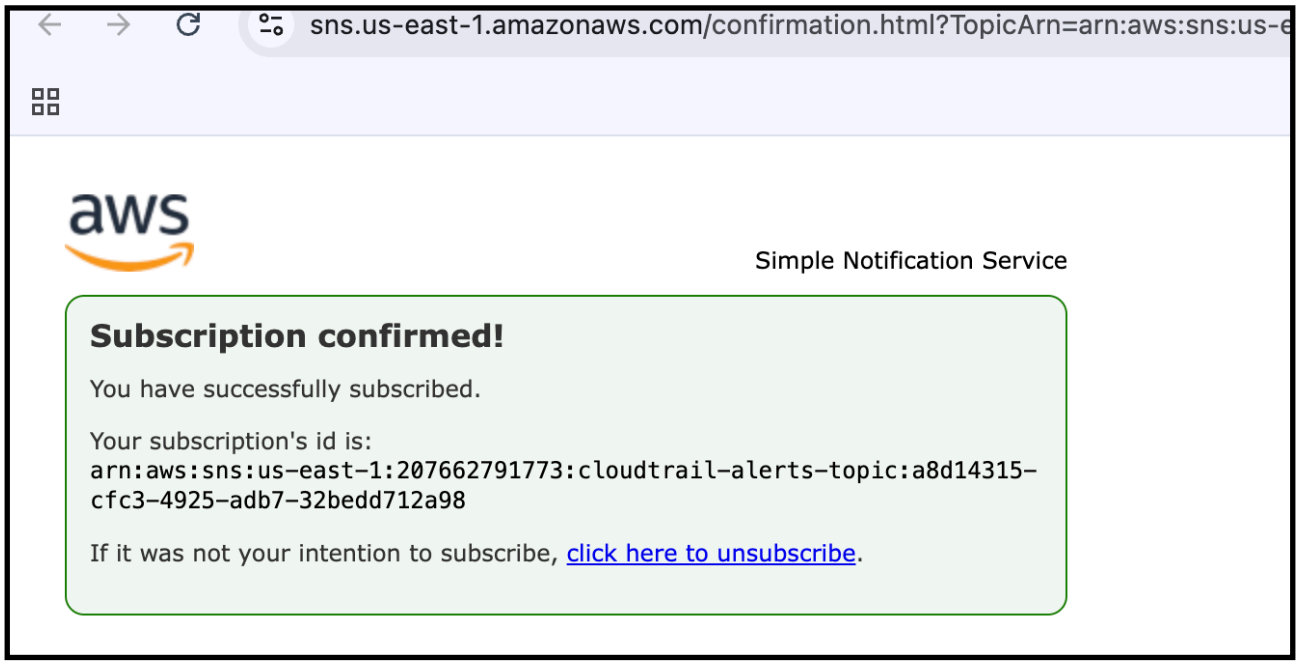
—> now give a subscription for that sns topic
—> by this we can give subscription for which sms, emails etc to get the notifications



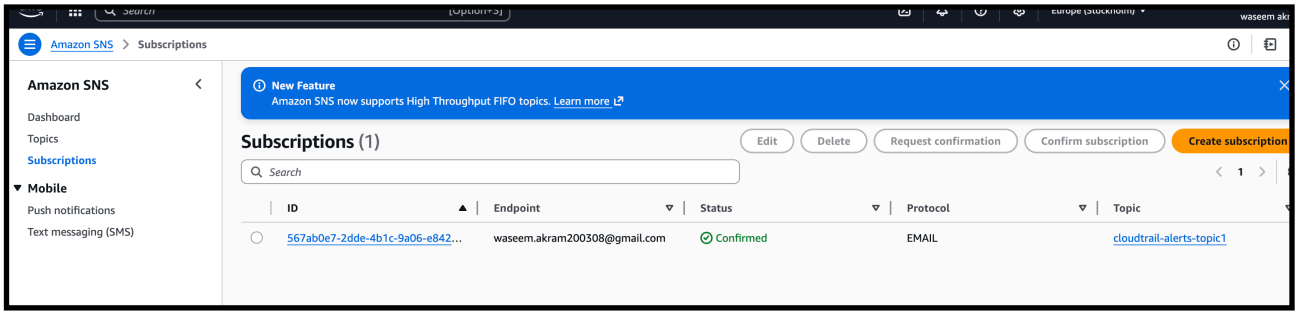
—> for now we are using email
—> and adding the details of the email name to get subscription request



—> after that check the gmail of the email we given we have to accept the subcription and check th status

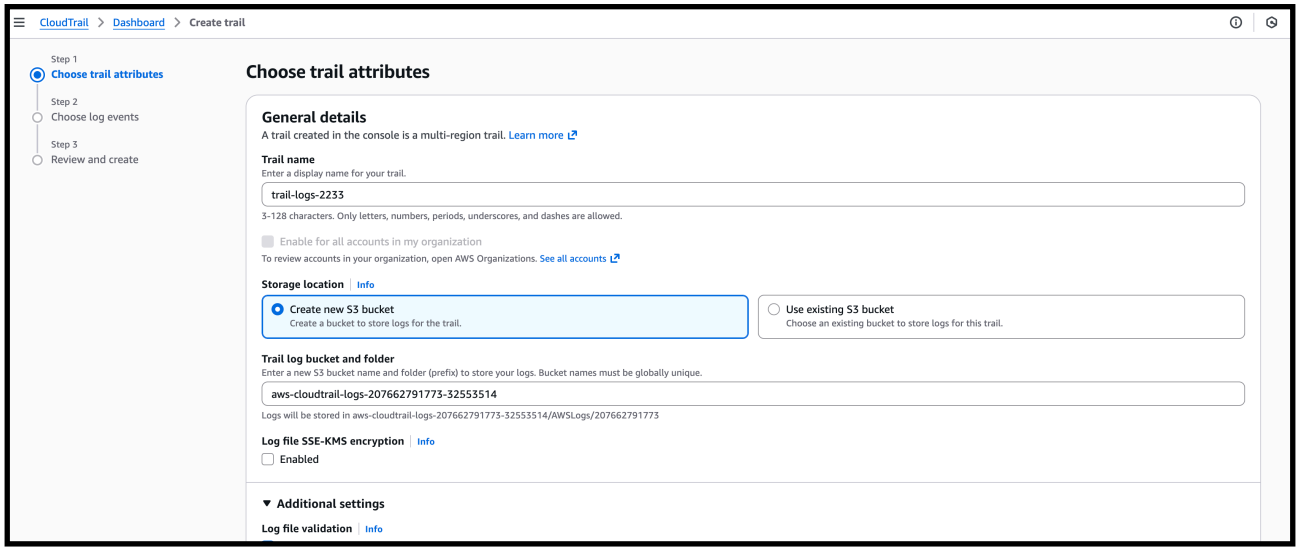


—> it will show the status as confirm if we subscribe

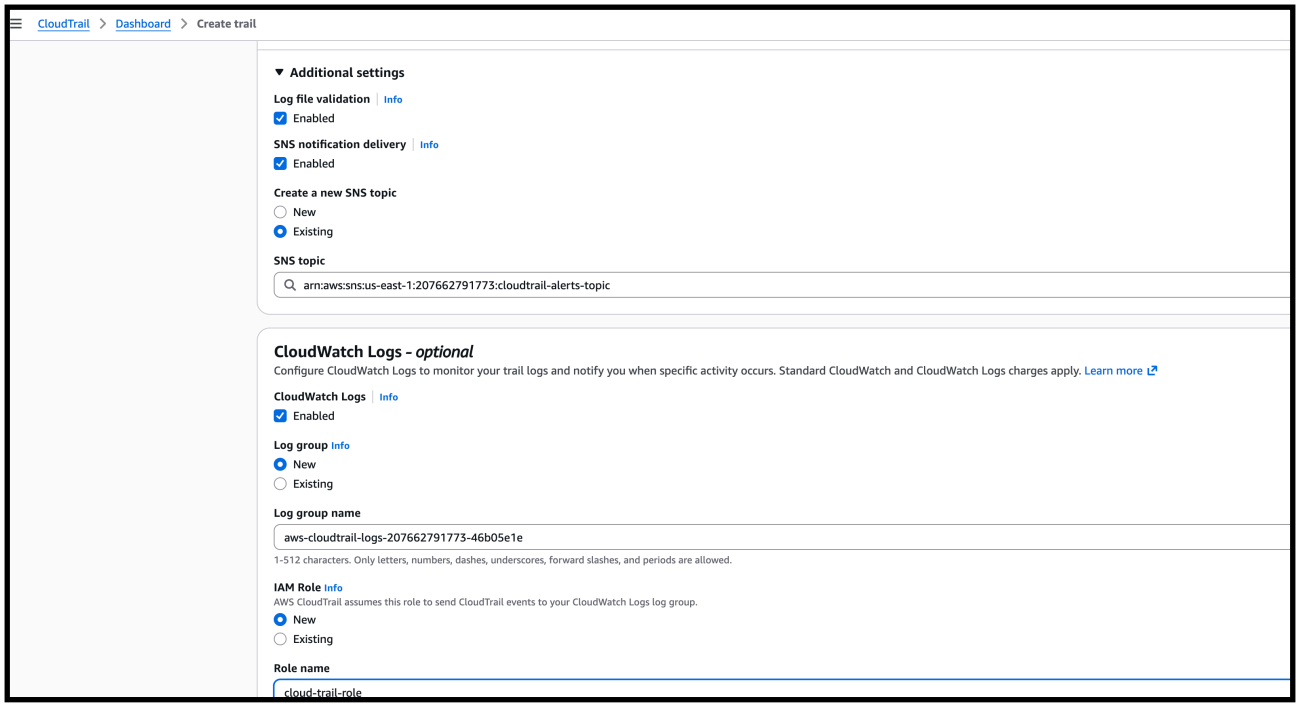


—> now we have to create a cloud trail and attach the sns for it

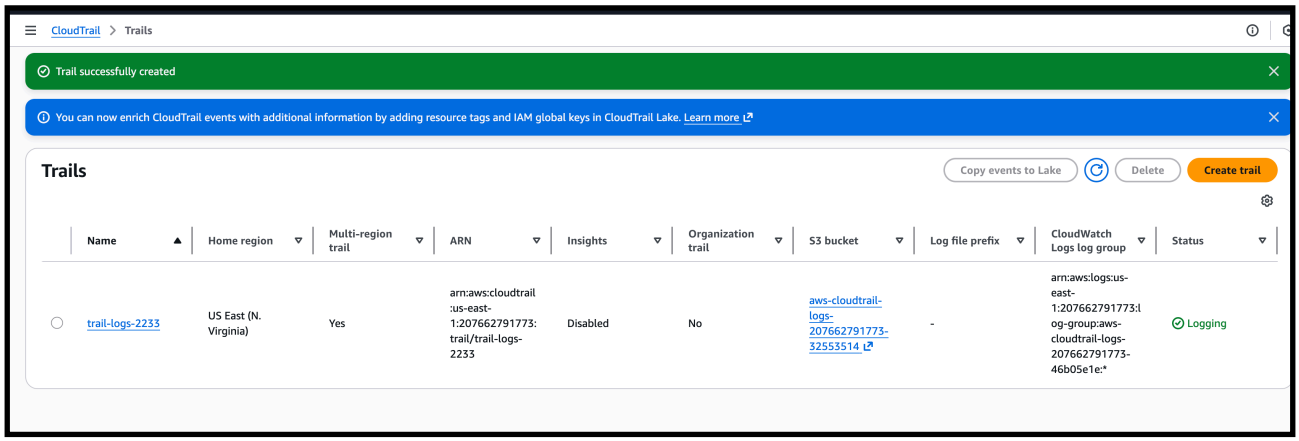
—> give a s3 bucket to create it



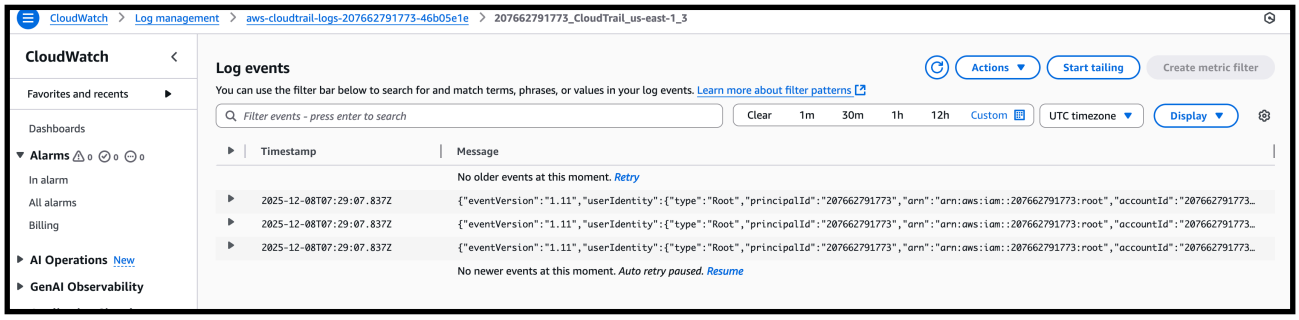
—> we can se the sns we created is available now select it



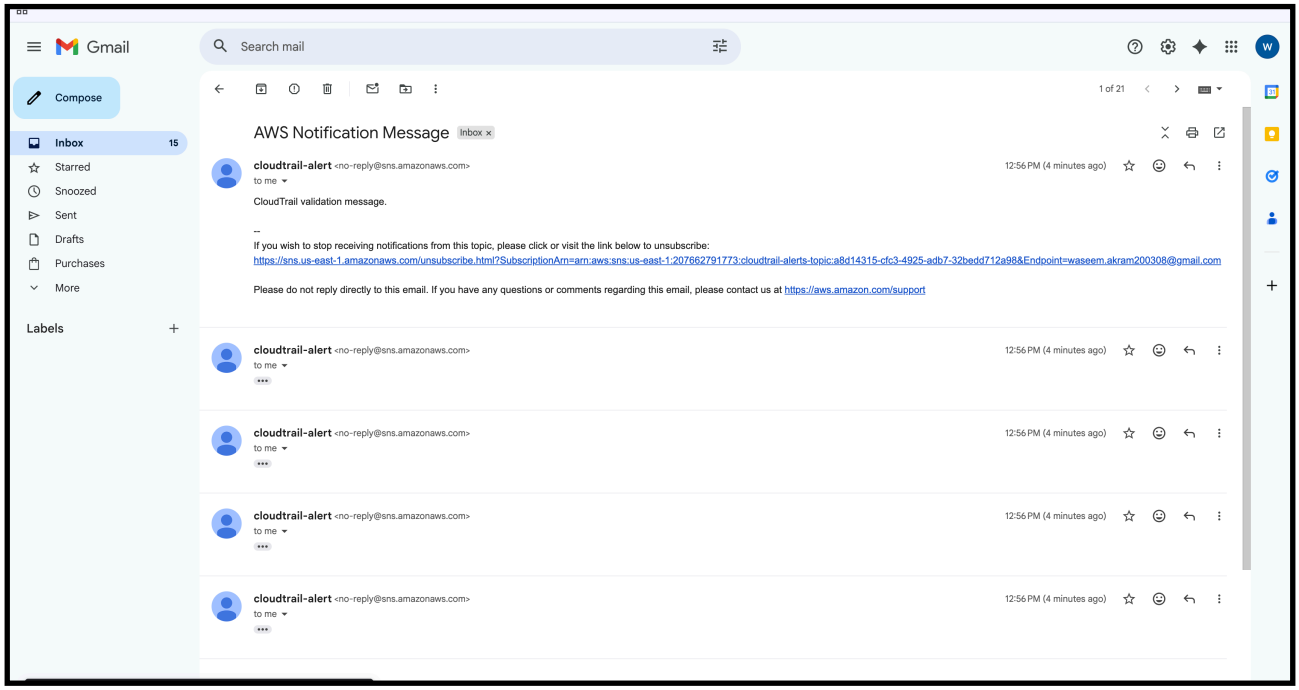
—> and created the cloudtail with sns



—> this are our cloud watch logs

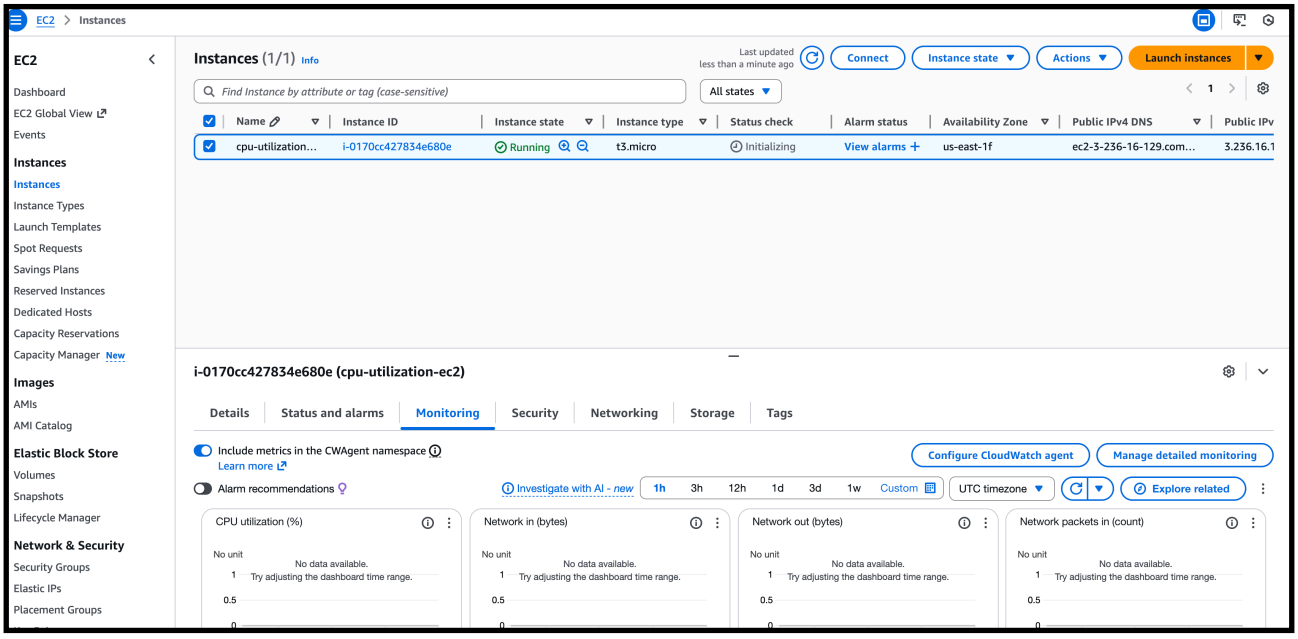


—> this are cloud trail logs in aur email

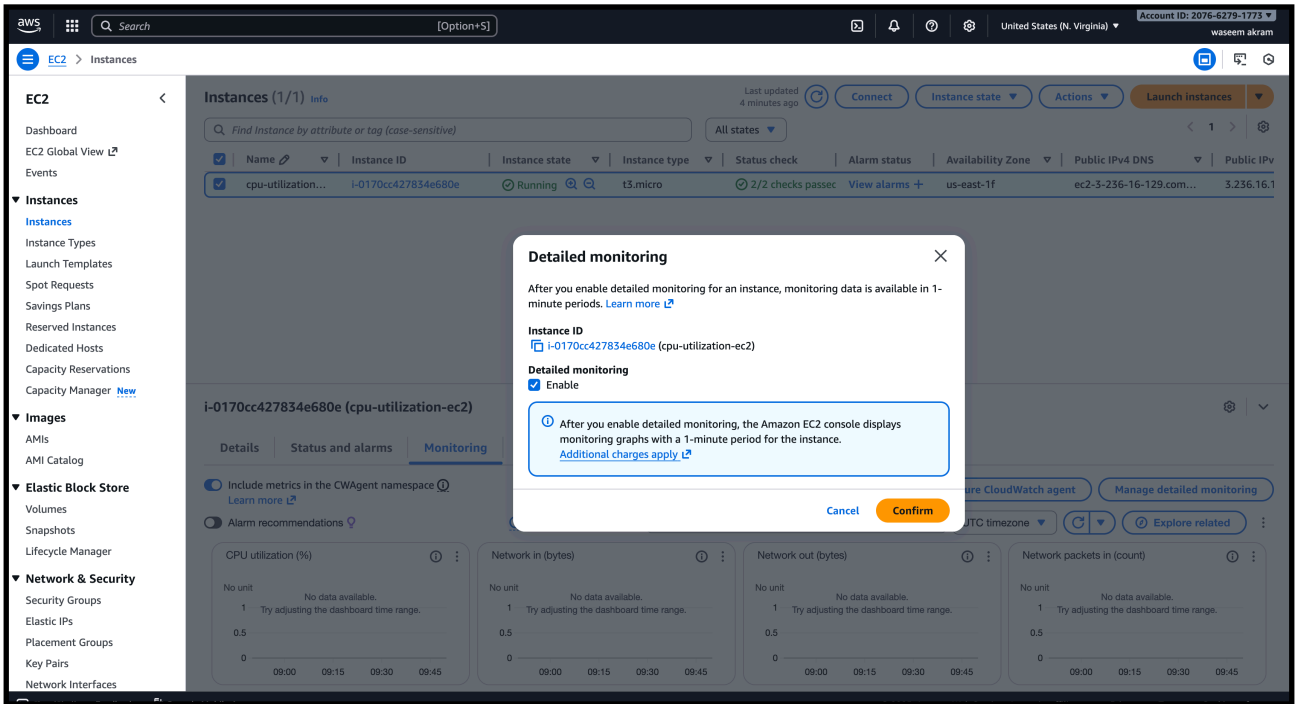


3) Configure cloud watch monitoring and record the cpu utilization and other metrics of ec2

AWS already records EC2 metrics automatically.
You only need to **enable detailed monitoring + see the metrics**.
—> first we have to launch one instance and name it according to the task for cpu

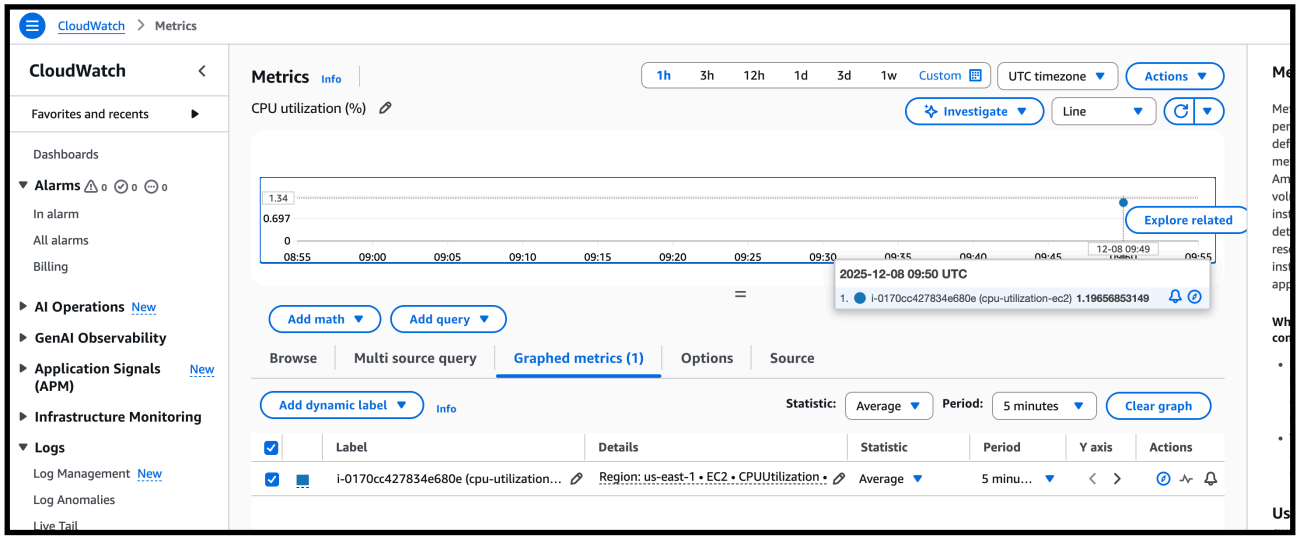


—> for monitoring the instance cpu utilization we have to click the instance—> monitoring —> and manage detailed monitoring
—> now we have ton enable the monitoring for this instance



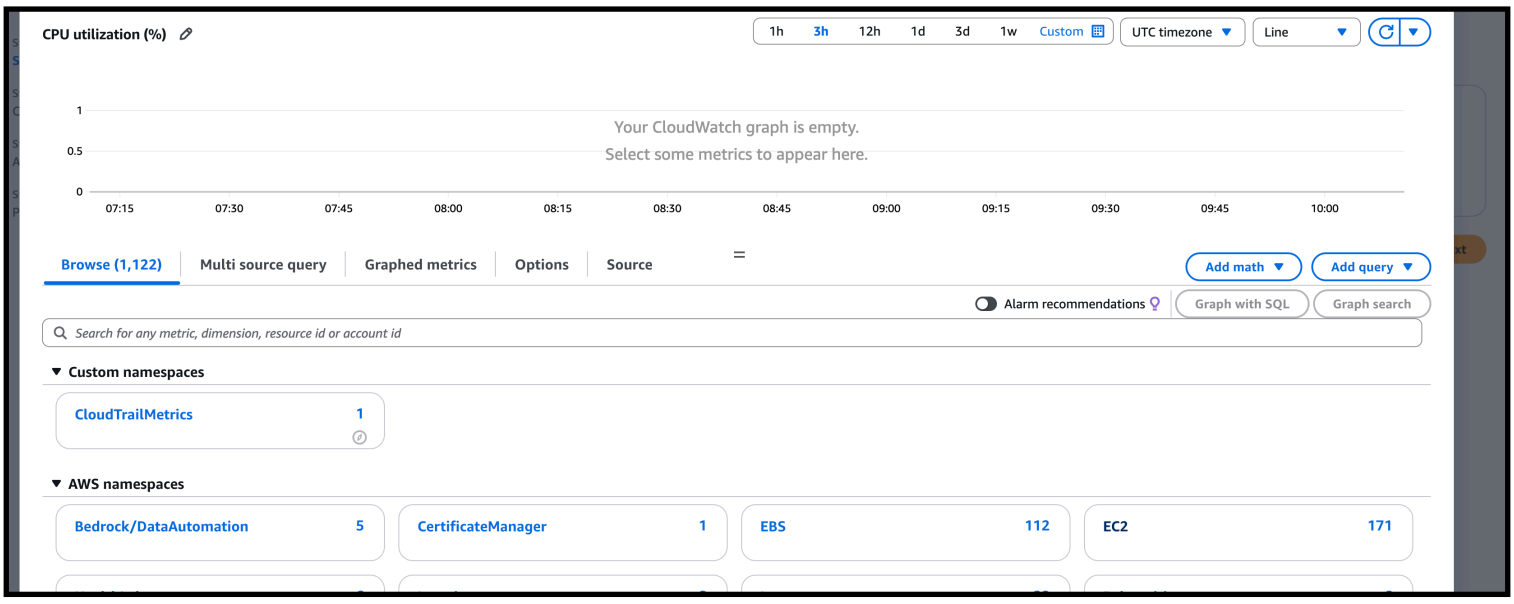
—>now we have to add some load to view the cpu utlization

—> after adding any service inside ec2 check the monitoring of the cpu

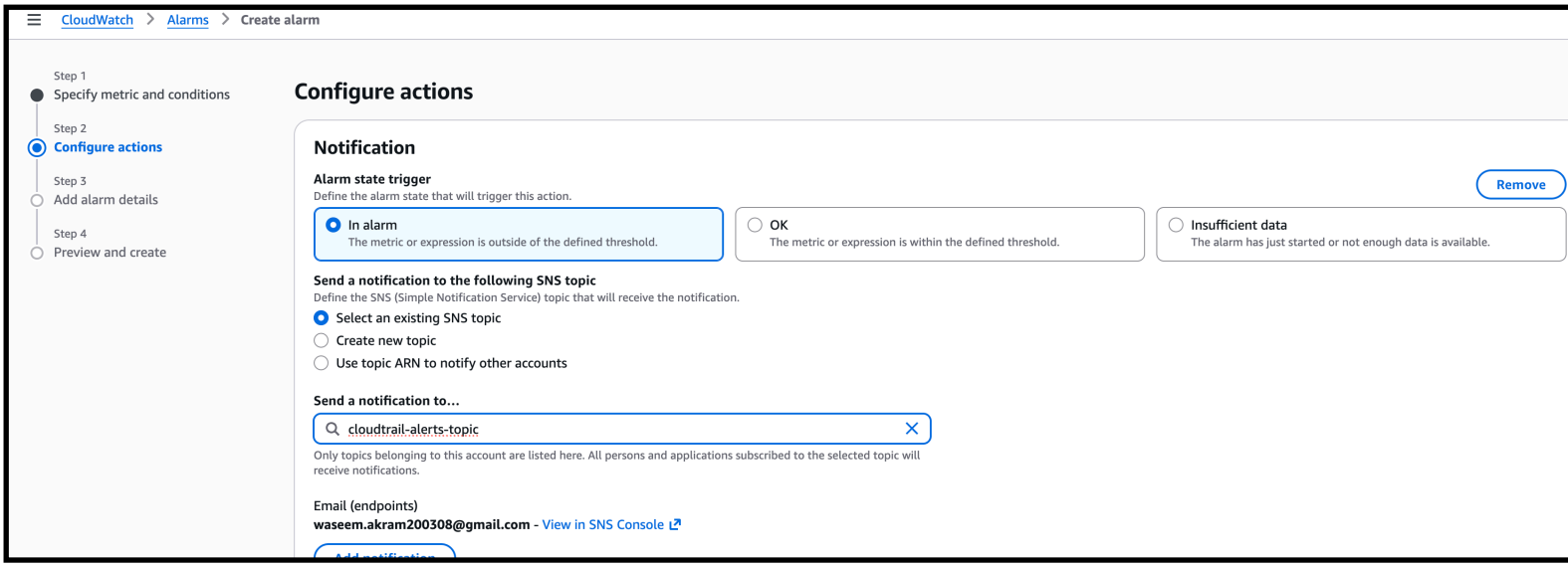


4) Create one alarm to send alert to email if the cpu utilization is more than 70 percent

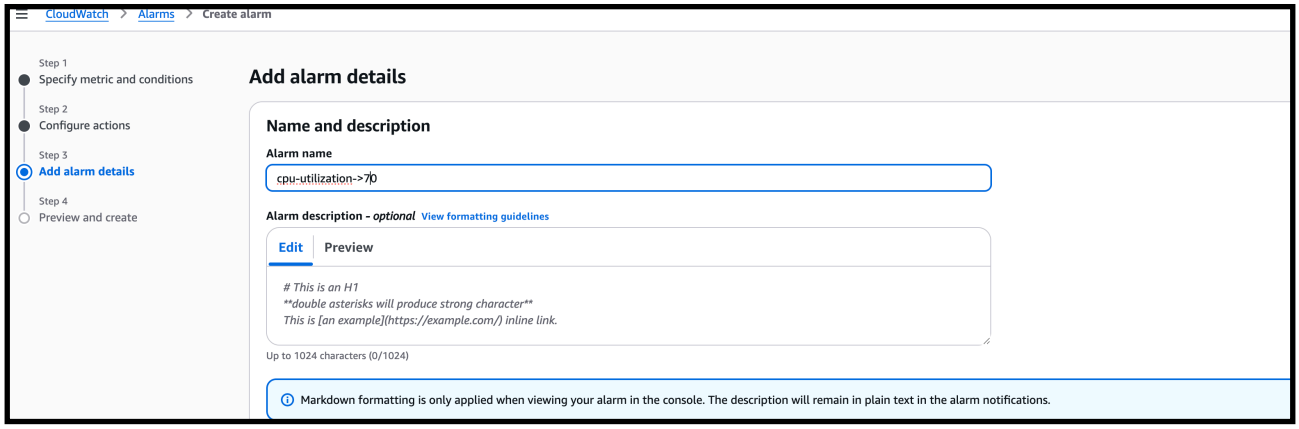
—> go to the alarm —> create alarm
—> select matric —> ec2



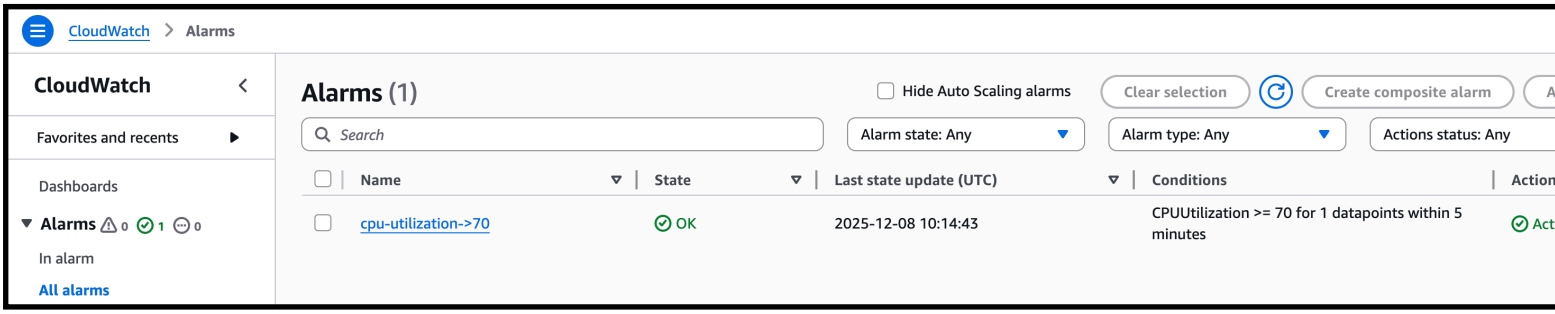
—> set at inalarm for getting sns notification from gmail



—> name the alarm according to to task “cpu”
—> click next

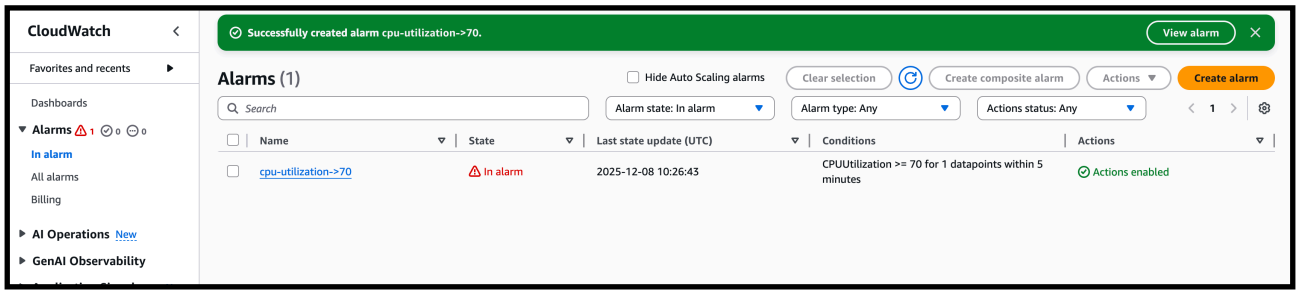


—> after clicking create . aur alarm for cpu utilization is successfully created

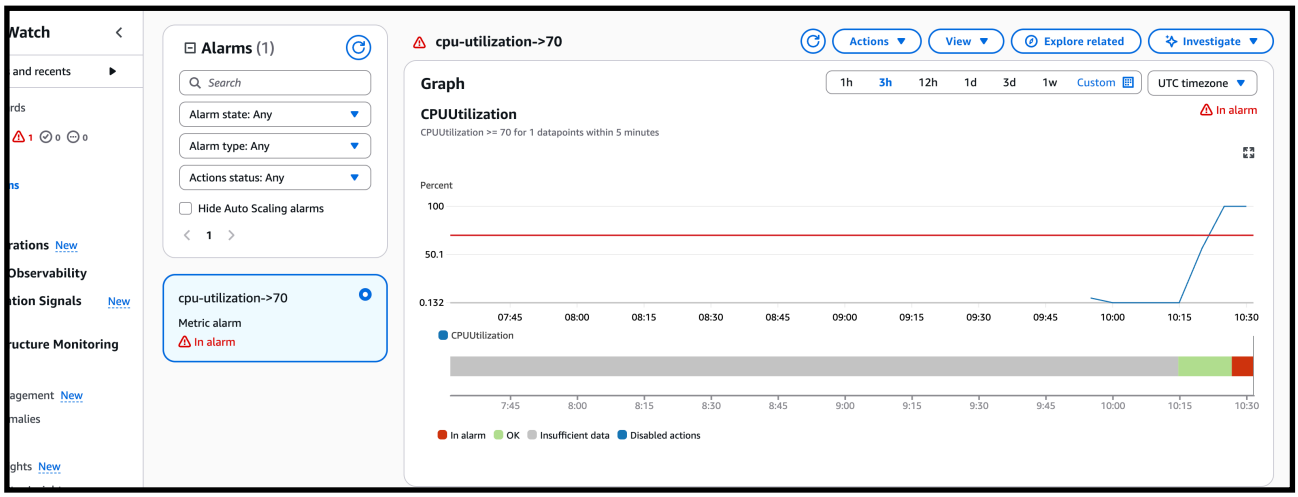


—> connect your ec2 instance and add some fake load
—> aur ec2 load is less than 70 so to add some fake load we can use “Yes > /dev/null & “ to add some fake load to ec2

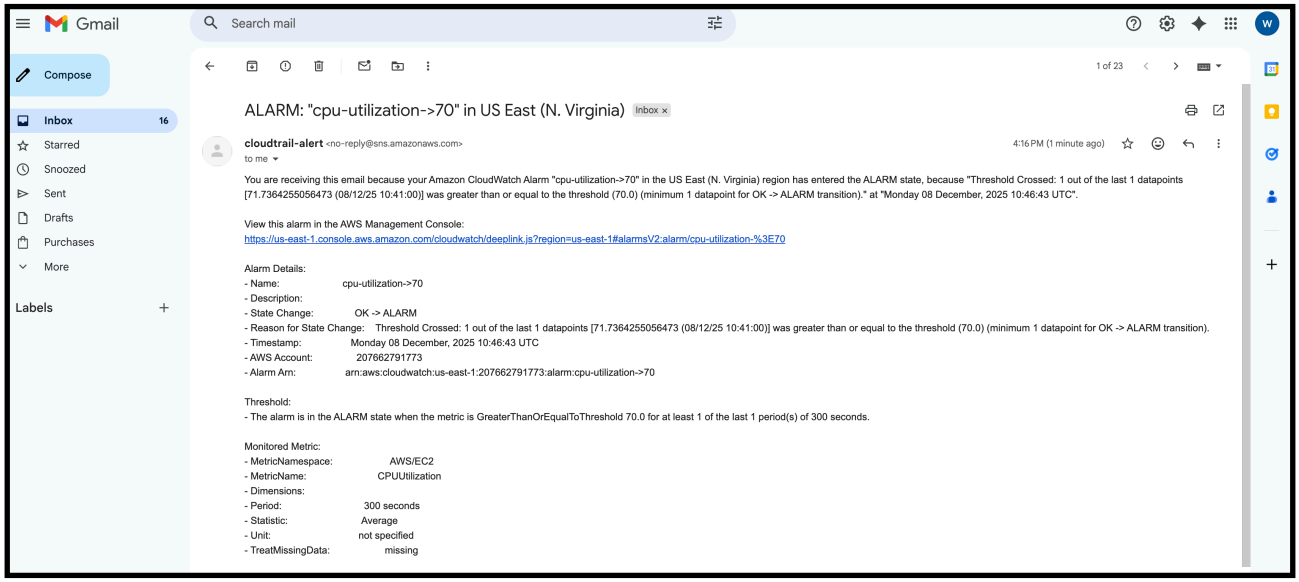
```
Complete!
[ec2-user@ip-172-31-75-244 ~]$ yes > /dev/null &
yes > /dev/null &
yes > /dev/null &
yes > /dev/null &
[1] 26377
[2] 26378
[3] 26379
[4] 26380
[ec2-user@ip-172-31-75-244 ~]$
```



—> now check the cloud watch of the matric of cpu utilization
—> we can see the load is more than 70 now we will be getting a notification alarm in our gmail



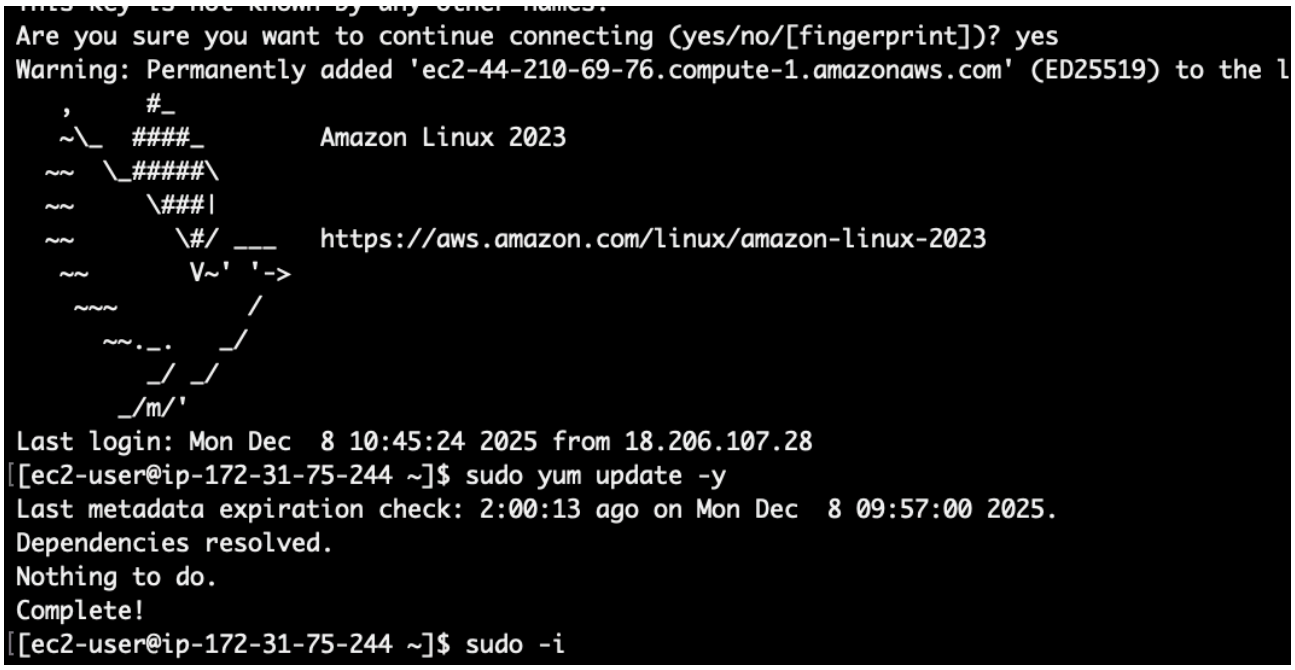
—> I have opened the email send by aws of cpu utilization more than 70



5) Create Dashboard and monitor tomcat service wether it is running or not and send the alert.

Follow all the below steps for the task

1. Launch an EC2 instance



—> Install Java and Tomcat

```
[root@ip-172-31-75-244 ~]# sudo dnf install -y java-17-amazon-corretto
Last metadata expiration check: 2:03:08 ago on Mon Dec  8 09:57:00 2025.
Dependencies resolved.
================================================================================
 Package                                                        Architecture
================================================================================
Installing:
 java-17-amazon-corretto                                        x86_64
Installing dependencies:
 alsa-lib                                                       x86_64
 cairo                                                          x86_64
 dejavu-sans-fonts                                              noarch
 dejavu-sans-mono-fonts                                         noarch
 dejavu-serif-fonts                                             noarch
```

-> sudo dnf install -y tomcat
./startup.sh to start the tomcat

```
Complete!
[root@ip-172-31-75-244 ~]# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.112/bin/apache-tomcat-9.0.112.tar.gz
--2025-12-08 12:01:09--  https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.112/bin/apache-tomcat-9.0.112.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13043951 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.112.tar.gz'

apache-tomcat-9.0.112.tar.gz                        100%[================================================>]

2025-12-08 12:01:09 (186 MB/s) - 'apache-tomcat-9.0.112.tar.gz' saved [13043951/13043951]

[root@ip-172-31-75-244 ~]# ls
apache-tomcat-9.0.112.tar.gz
[root@ip-172-31-75-244 ~]# mv apache-tomcat-9.0.112.tar.gz /opt
[root@ip-172-31-75-244 ~]# ls
[root@ip-172-31-75-244 ~]# cd opt
-bash: cd: opt: No such file or directory
[root@ip-172-31-75-244 ~]# cd /opt
[root@ip-172-31-75-244 opt]# ls
apache-tomcat-9.0.112.tar.gz  aws
[root@ip-172-31-75-244 opt]# tar xvf apache-tomcat-9.0.112.tar.gz
apache-tomcat-9.0.112/conf/
apache-tomcat-9.0.112/conf/catalina.policy
```

```
[root@ip-172-31-75-244 apache-tomcat-9.0.112]# cd bin
[root@ip-172-31-75-244 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.112
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.112
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.112/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.112/bin/bootstrap.jar:/opt/apache-tomcat-9.0.112/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

2) -> Create Tomcat monitoring script

```
#!/bin/bash

# Check if Tomcat is running
if pgrep -f tomcat > /dev/null
then
    STATUS=1
else
    STATUS=0
fi

# Send custom metric to CloudWatch
aws cloudwatch put-metric-data \
    --metric-name tomcat_status \
    --namespace "CustomTomcat" \
    --value $STATUS \
    --region us-east-1
```

```
[[root@ip-172-31-75-244 bin]# cd
 [root@ip-172-31-75-244 ~]# sudo vi monitoring.bash
[[root@ip-172-31-75-244 ~]# chmod 755  monitoring.bash
```

--> Give execution permission
Chmod 755 for this script file

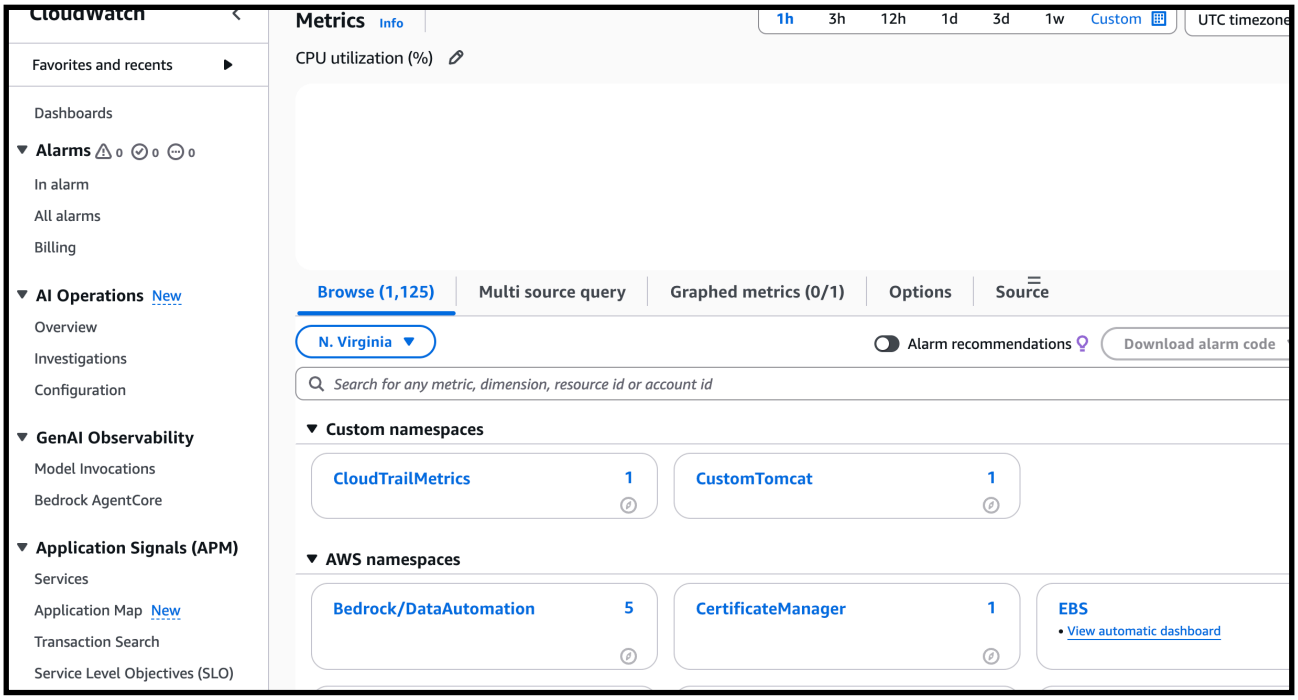
3) aws configure

-> inside your ec2 give aws configure to access the root user of your account

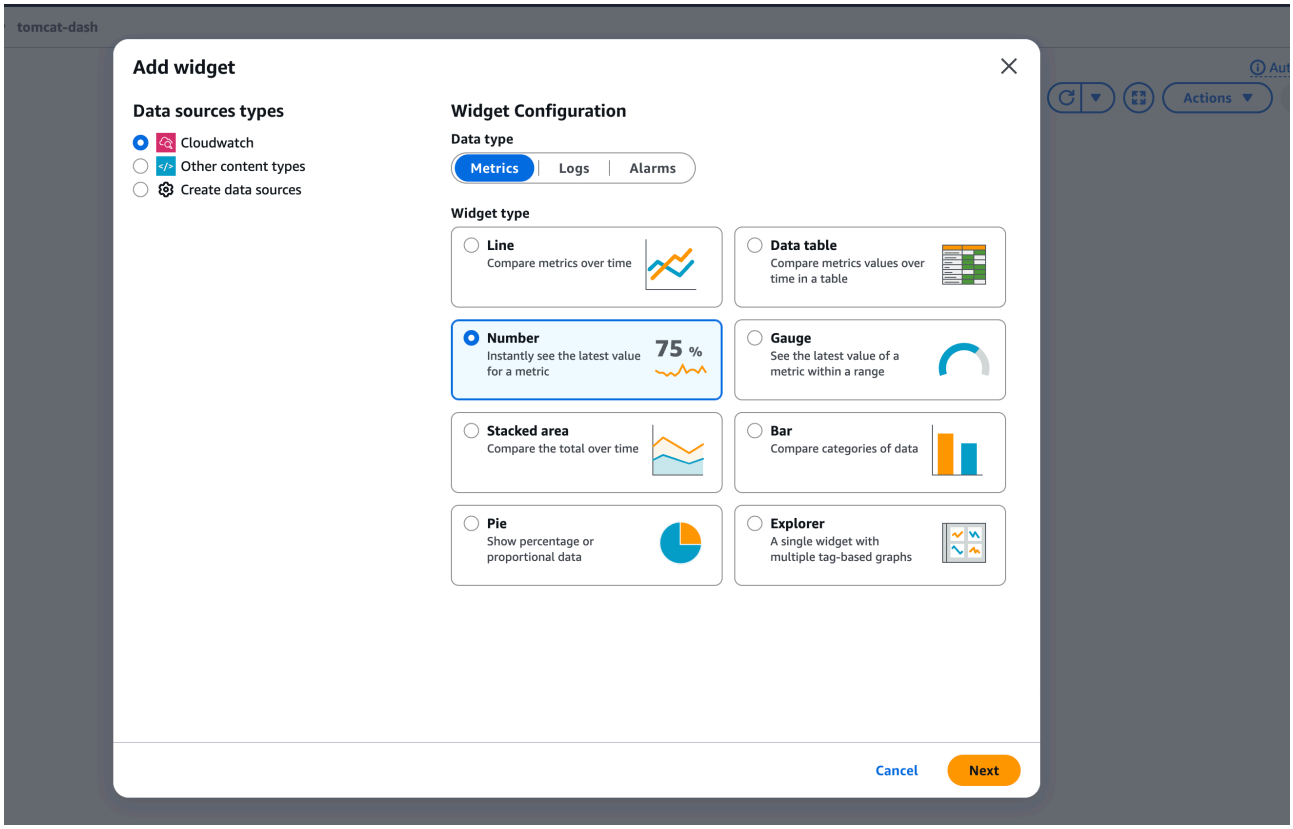
```
[root@ip-172-31-75-244 ~]# sudo vi monitoring.bash
[[root@ip-172-31-75-244 ~]# chmod 755  monitoring.bash
 [root@ip-172-31-75-244 ~]# aws configure
[AWS Access Key ID [None]: AKIATAWNKXROZTWVRRMS
[AWS Secret Access Key [None]: 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
[Default region name [None]: us-east-1
[Default output format [None]: json
 [root@ip-172-31-75-244 ~]# ./monitoring.bash
```

-> after aws configuring we have to run the ./monitoring.bash script to. Run the script
-> this script will create a matric for the tomcat

-> next to check the matric we created by script
-> go to cloudwatch -> matrics -> browse -> custom tomcat
-> we are able to see the matric which we wanted to monitor



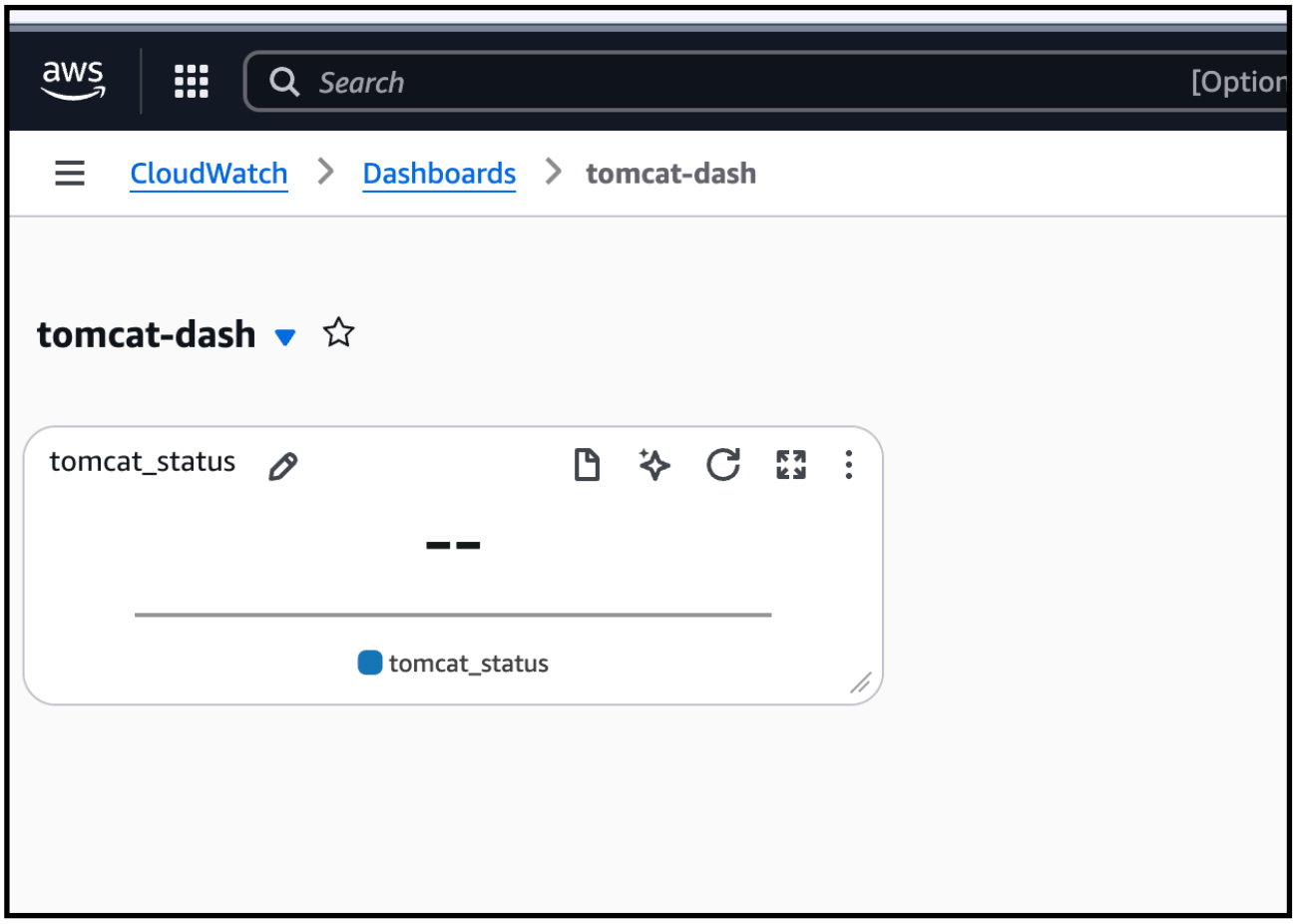
4)-> now create a dashboard
 -> select the custom tomcat
-> give numeric for better understanding



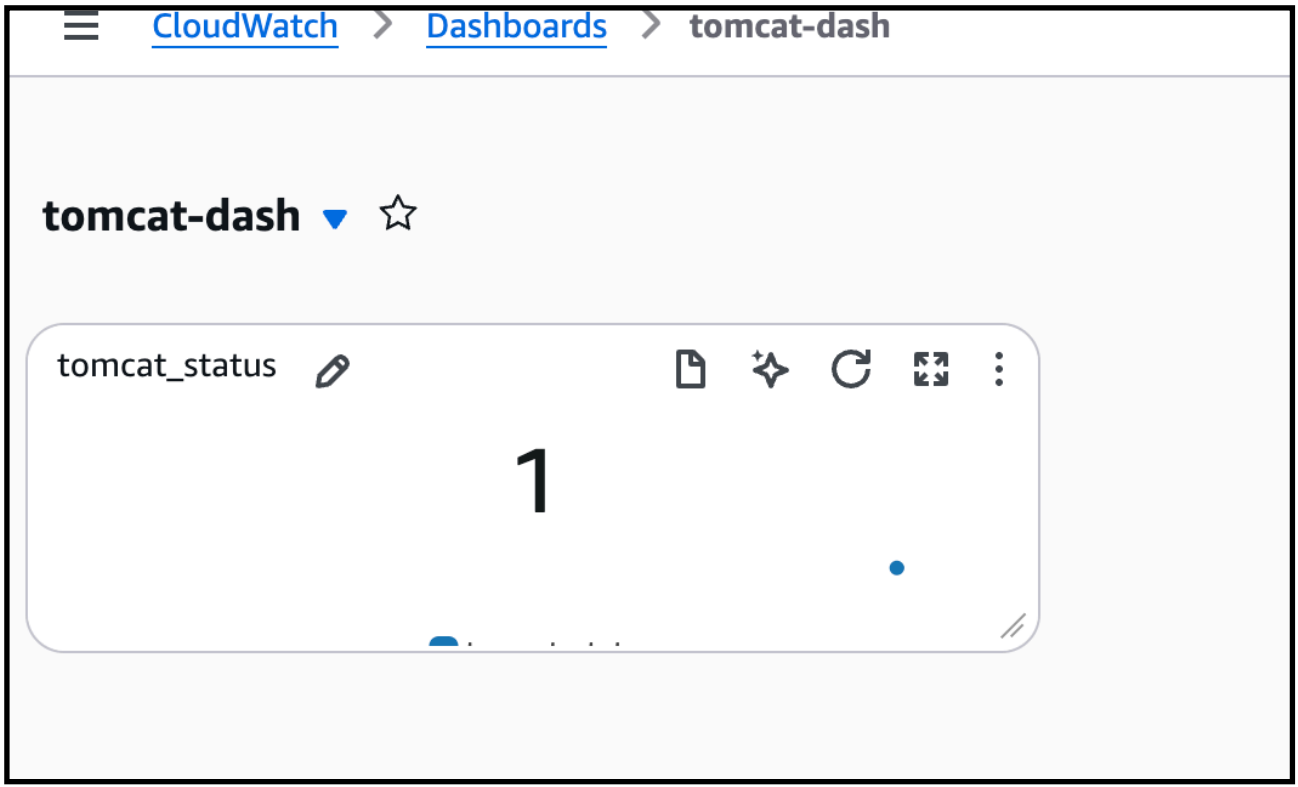
-> click next after selecting the metric

-> now we can see dashboard of the tomcat is available

-> now refresh after 1 min



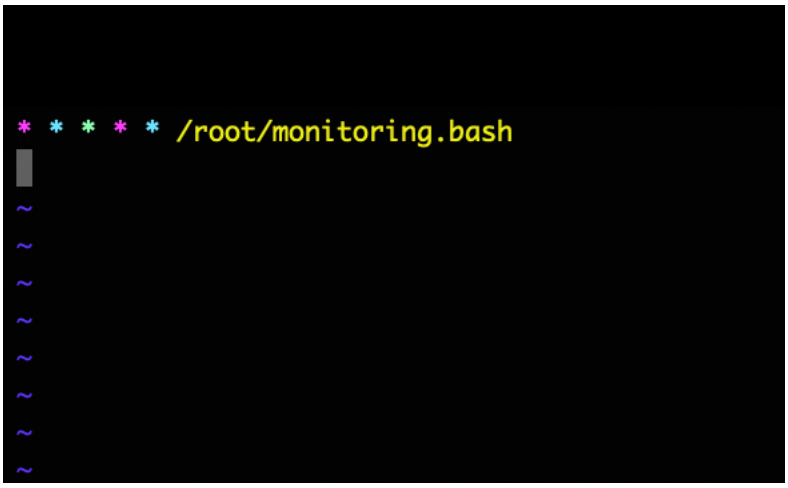
-> it will show 1 because tomcat it is active



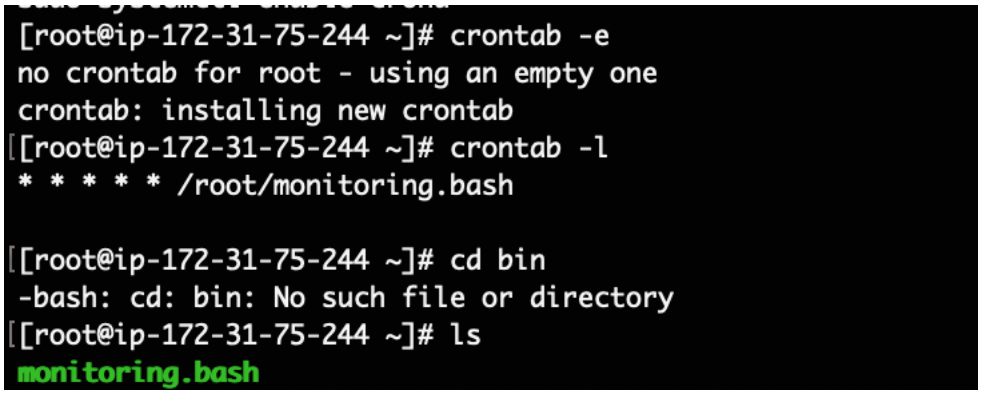
5)—>Configure cron job to send metric every minute

-> crontab -e to add the cron job

To send metric every minute

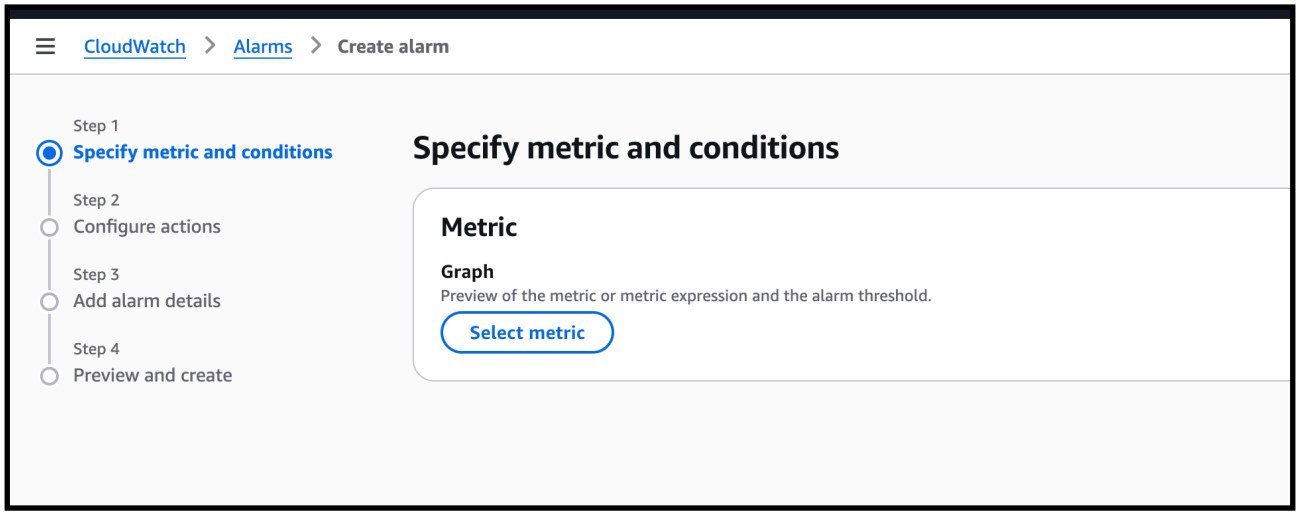


-> crontab -l for the list of cronjob

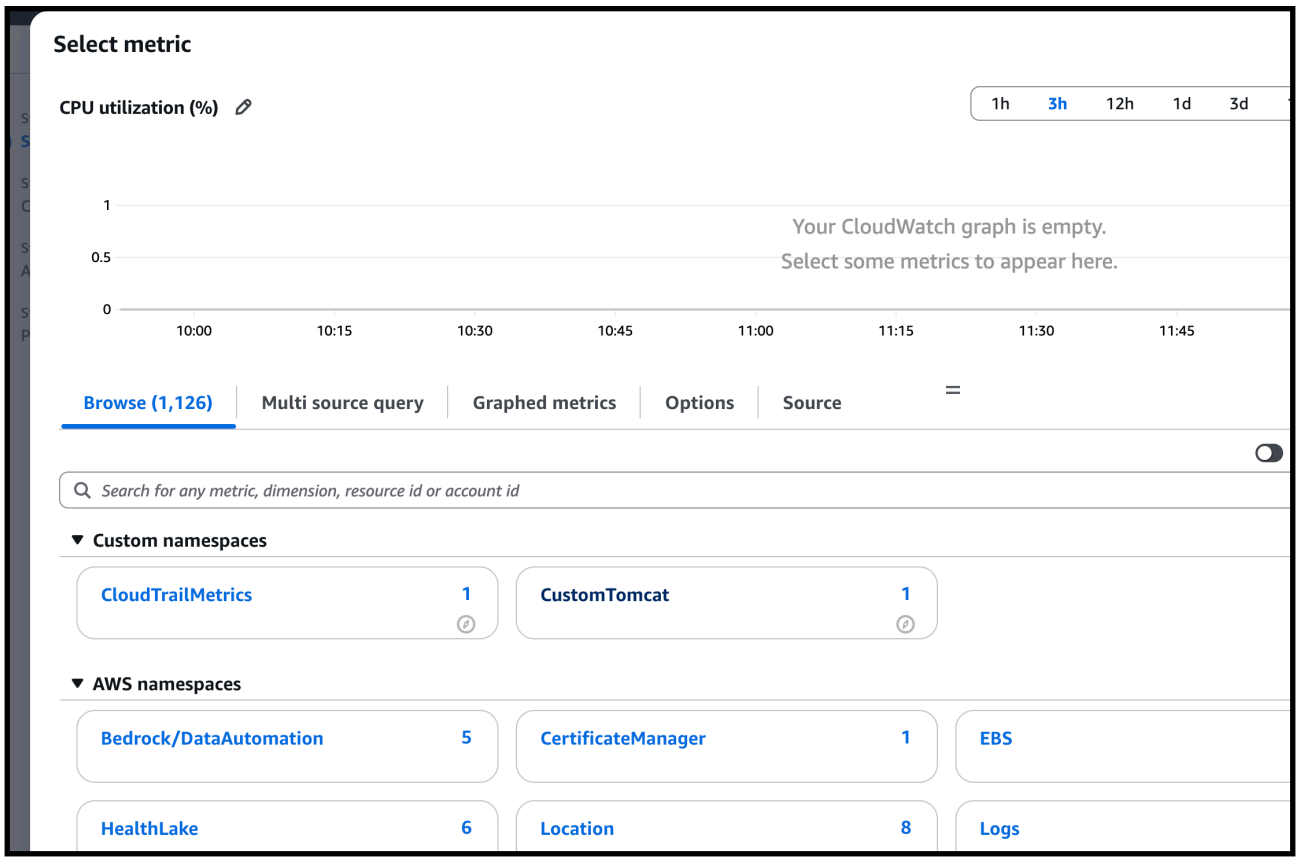


6) -> now lets create a alarm to get email when tomcat is shutdown 0 if tomcat status is zero ion dash board then send a email

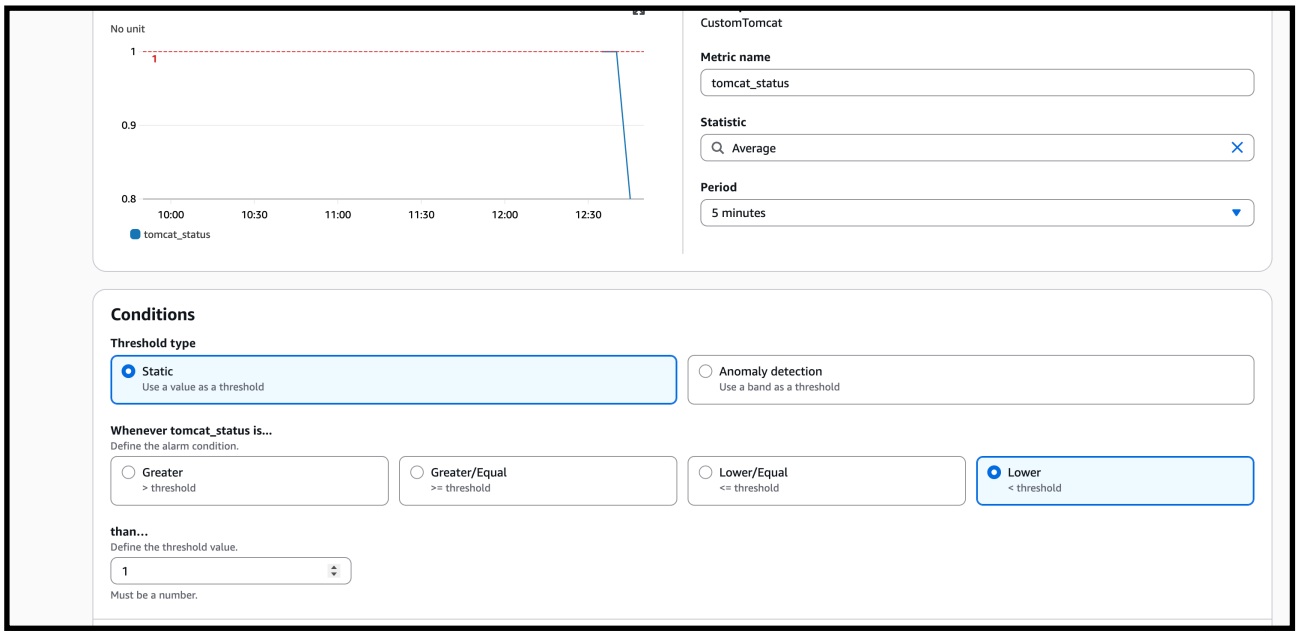
-> alarm -> create alarm -> select metric
-> we have to select the tomcat matric which we created



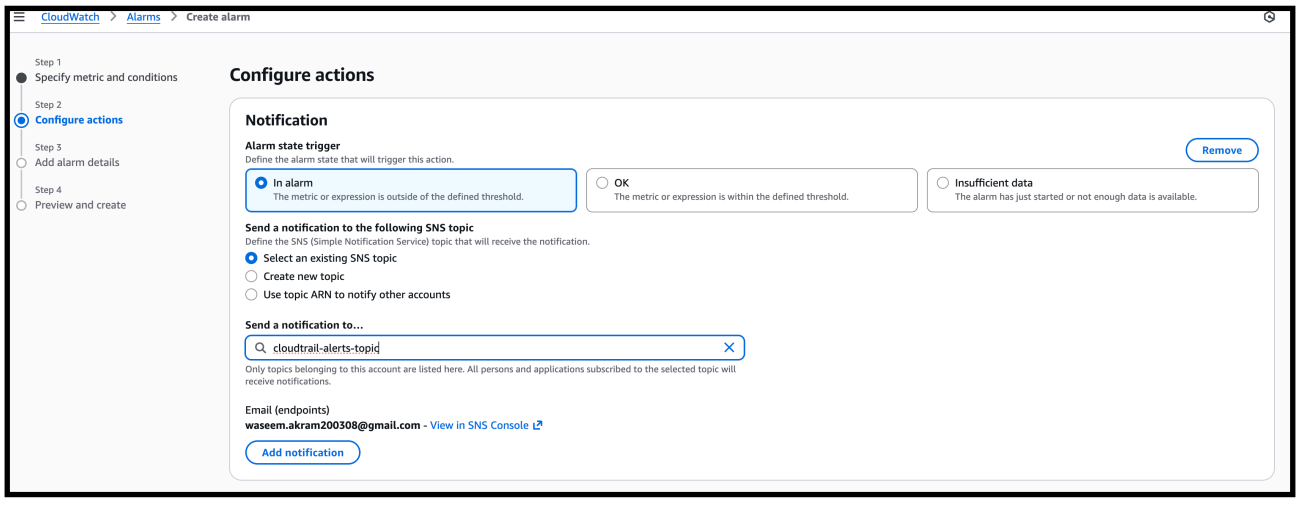
-> select the custom tomcat



-> now give the conditions of the alarm
-> we want to get alarm notification when our dashboard of tomcat is 0 (service stopped)

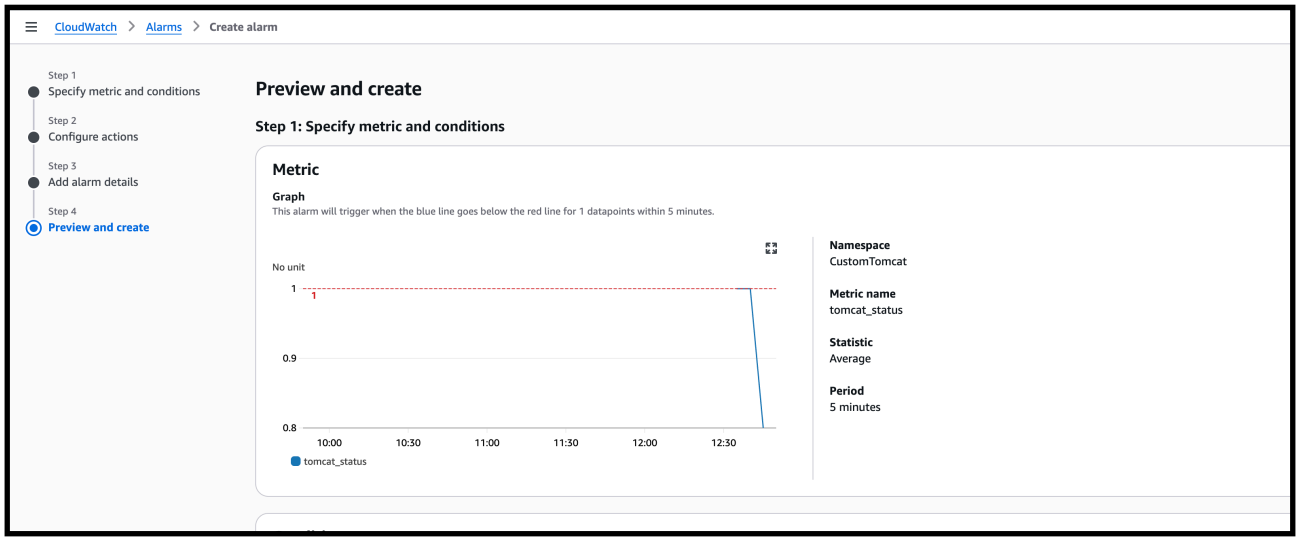


-> now in configure actions step
-> alarm start trigger -> select in alarm

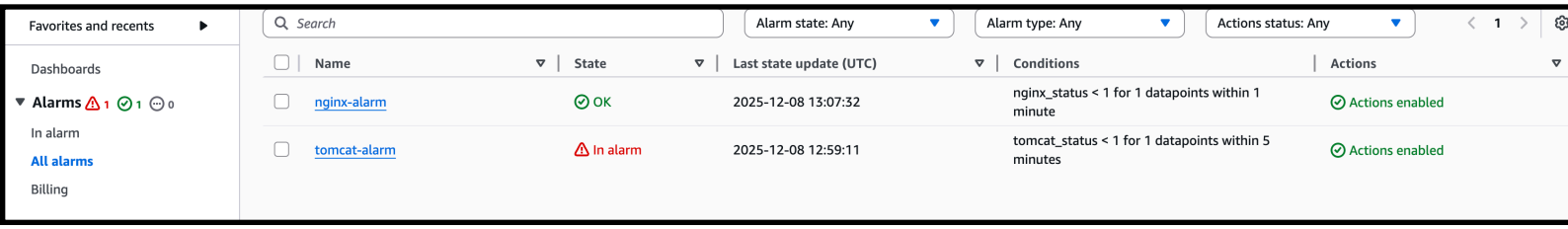


-> select the sns for the subscription which we created to get notification on particular email
-> so by this if the tomcat is stoped meaning dashboard is 0 then we will get a trigger alarm notification on our email which we attached to sns

-> click next and preview next

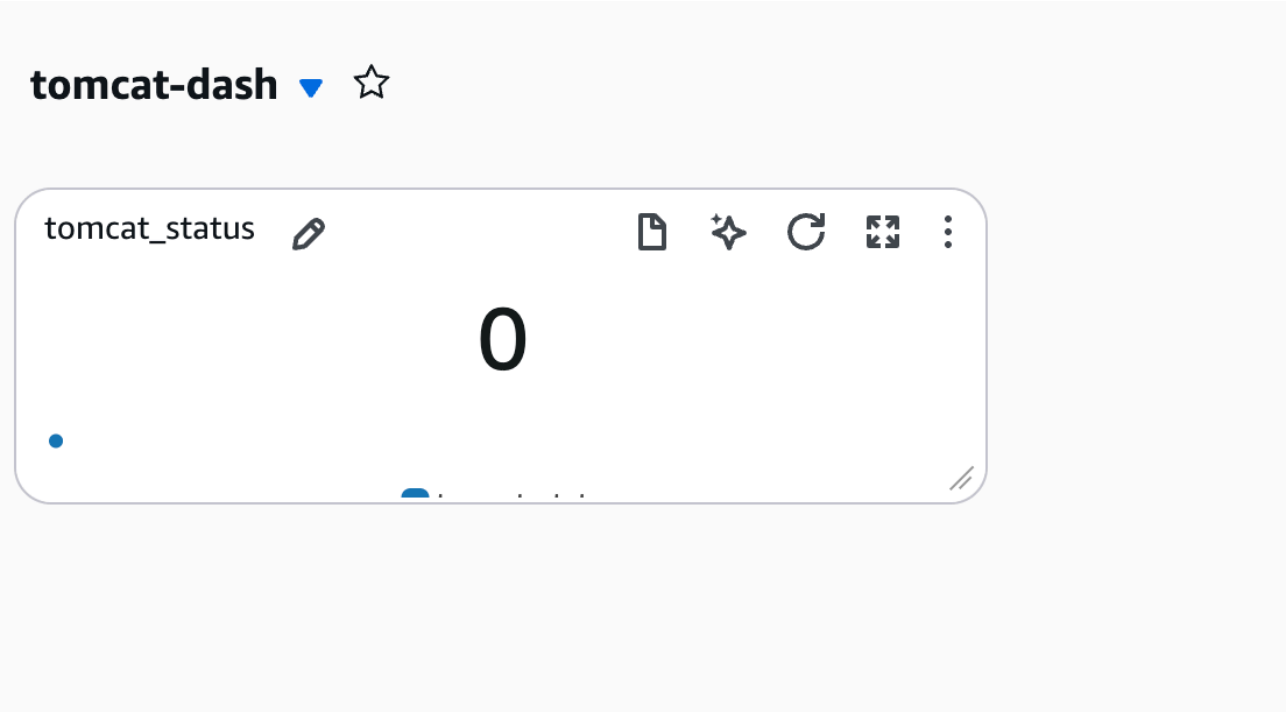


-> now to check the tomcat-alarm is created now it is in alarm state because the tomcat is shutdown

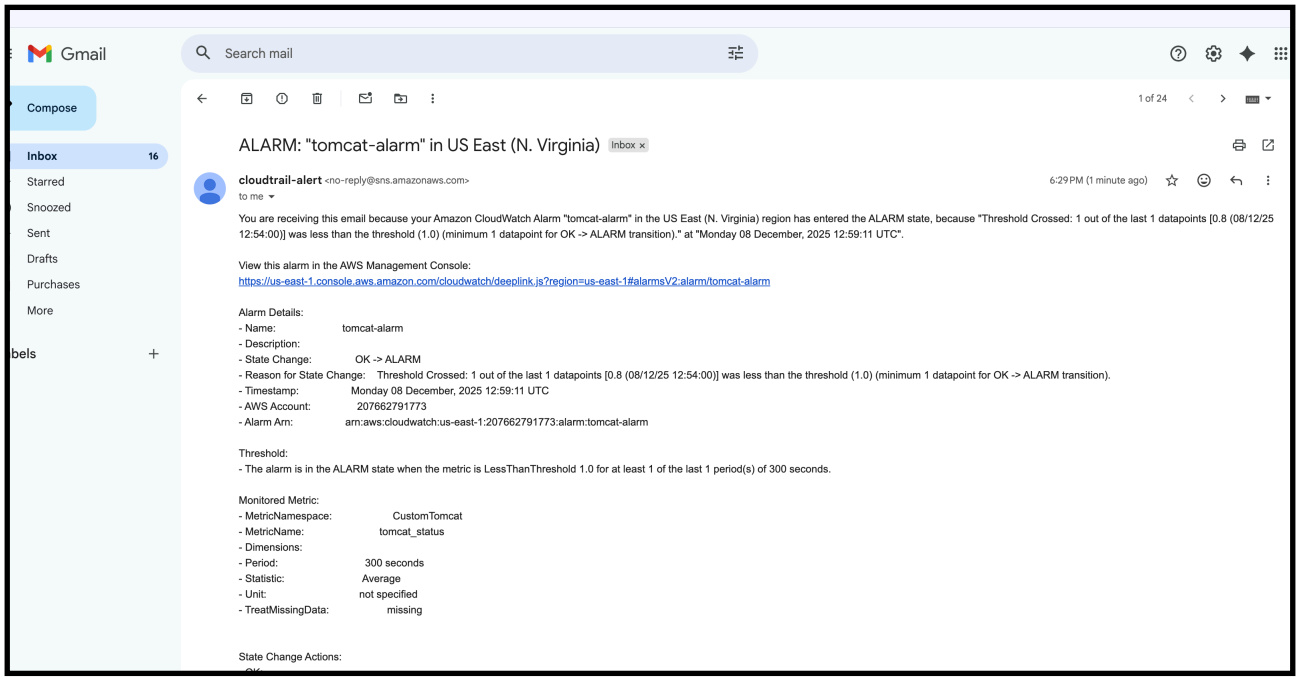


-> tomcat was shutdown

-> so we will be getting a alarm notification that tomcat dashboard is at 0

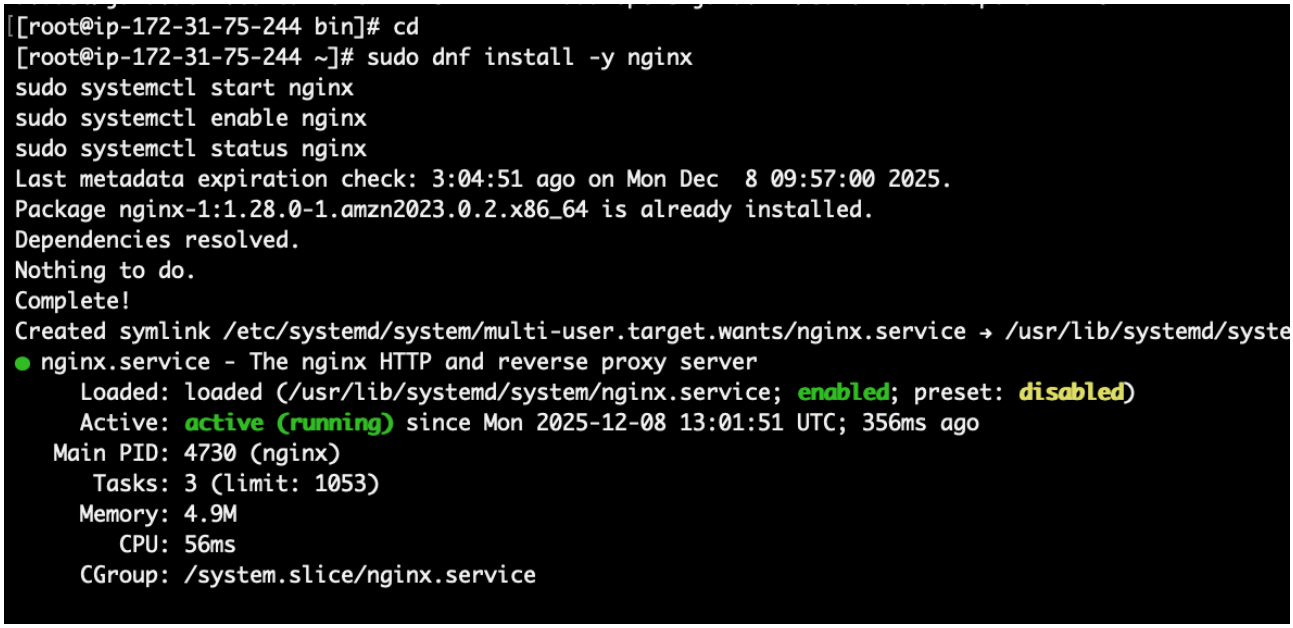


--> and we will be getting a alarm notification on aur gmail



6) Create Dashboard and monitor nginx service to send the alert if nginx is not running.

—-> now same as task 5 we have to do same with nginx
—> step by step follow
—> yum install nginx -y
Systemctl start nginx
Systemctl status nginx
—-> nginx is running



—— > after this we have too add a script for nginx to create its matric To monitore in aur aws cloudwatch

—-> create a file and add the below script to create a matric in aur cloudwatch

—> give execute permission to the file

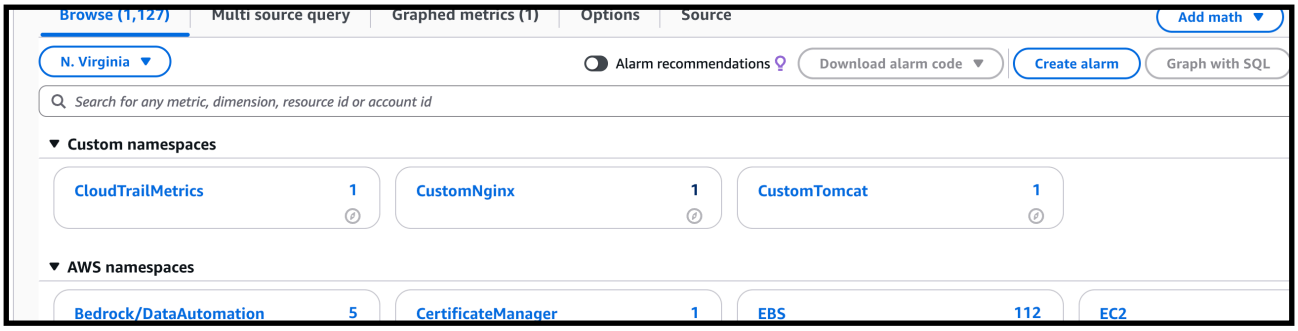
```
#!/bin/bash

# Check if NGINX is running
if pgrep -f nginx > /dev/null
then
    STATUS=1
else
    STATUS=0
fi

# Send custom metric to CloudWatch
aws cloudwatch put-metric-data \
    --metric-name nginx_status \
    --namespace "CustomNginx" \
    --value $STATUS \
    --region us-east-1
~
~
~
~
~
~
~
~
```

—> run the script ./monitoring_nginx.bash

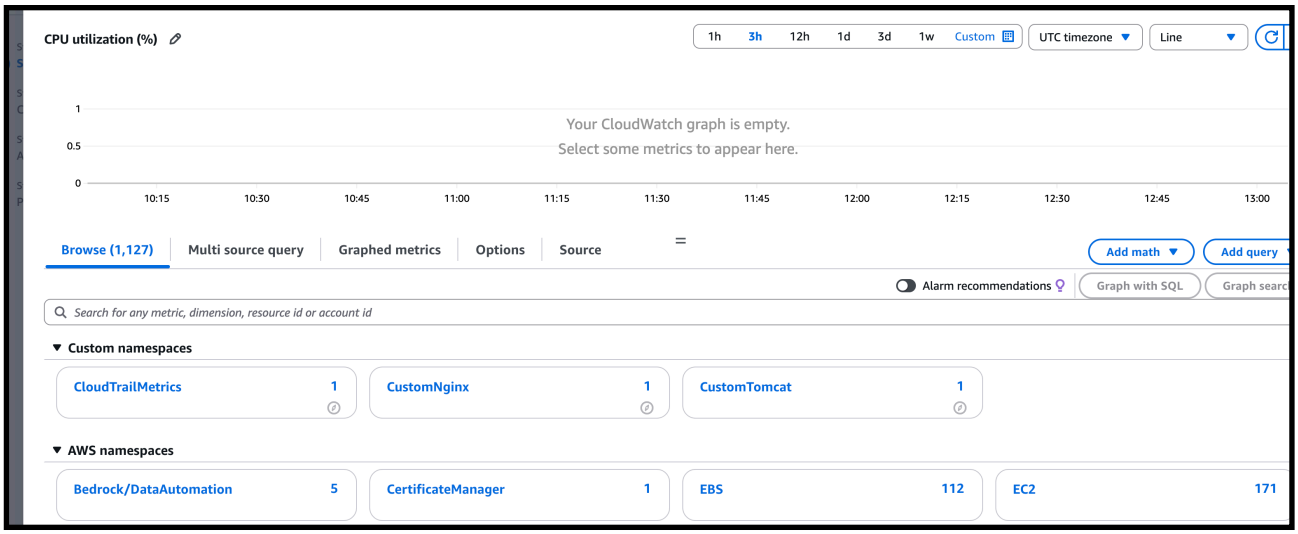
```
[root@ip-172-31-75-244 ~]# vi monitoring_nginx.bash
[root@ip-172-31-75-244 ~]# chmod +x monitoring_nginx.bash
[root@ip-172-31-75-244 ~]# ./monitoring_nginx.bash
```

—-> now go and check the matric —> browse

—> we are able to see there is customnginx matric created by the script

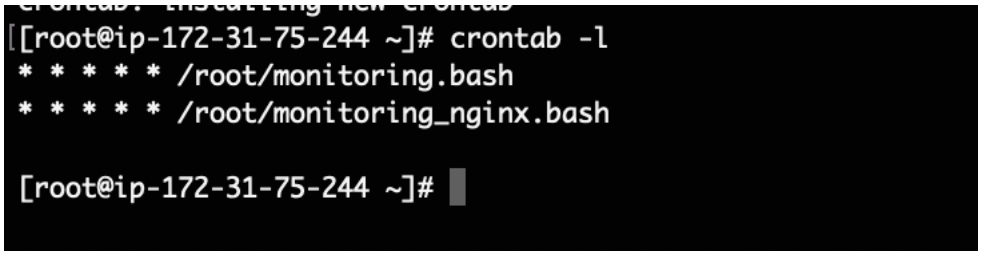


—-> now creates a dash board for the nginx monitoring

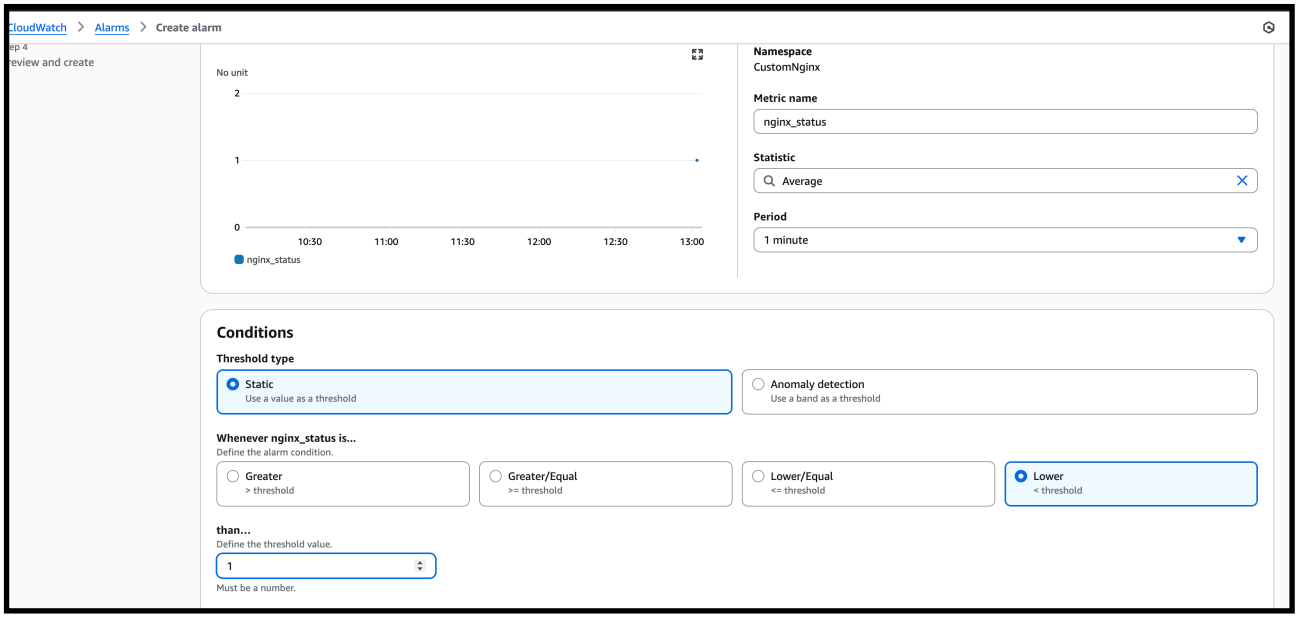


—-> dashboard is created for nginx

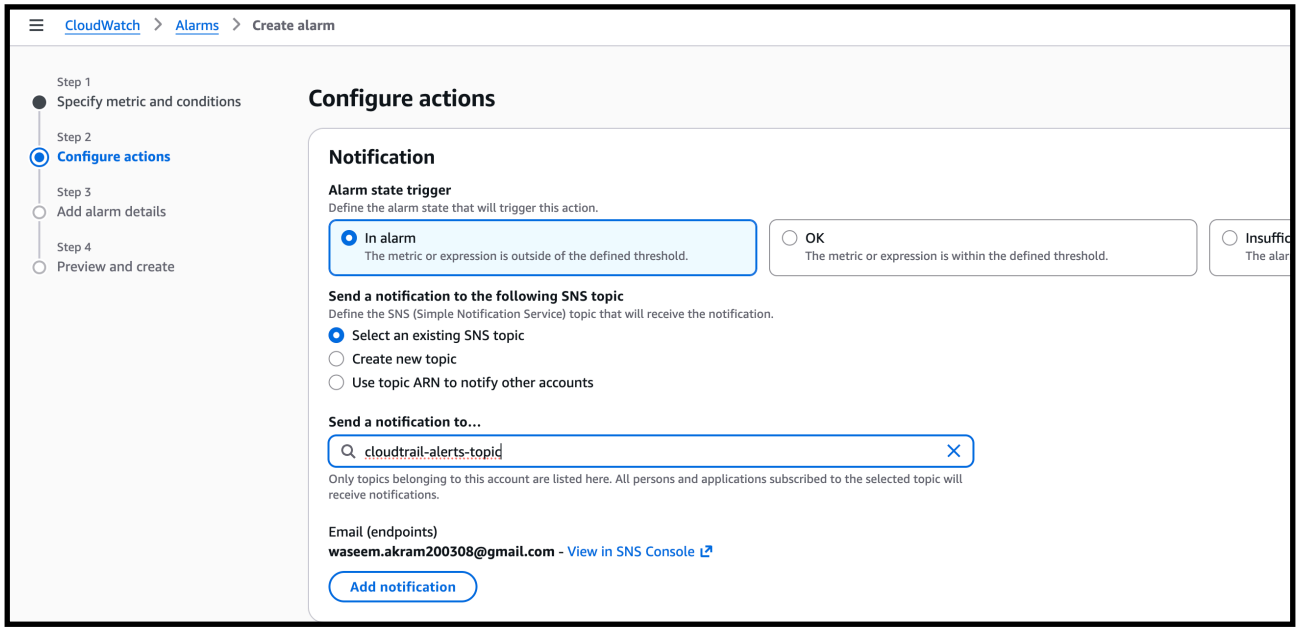
—-> give the same crontab rule for nginx also to check the monitoring



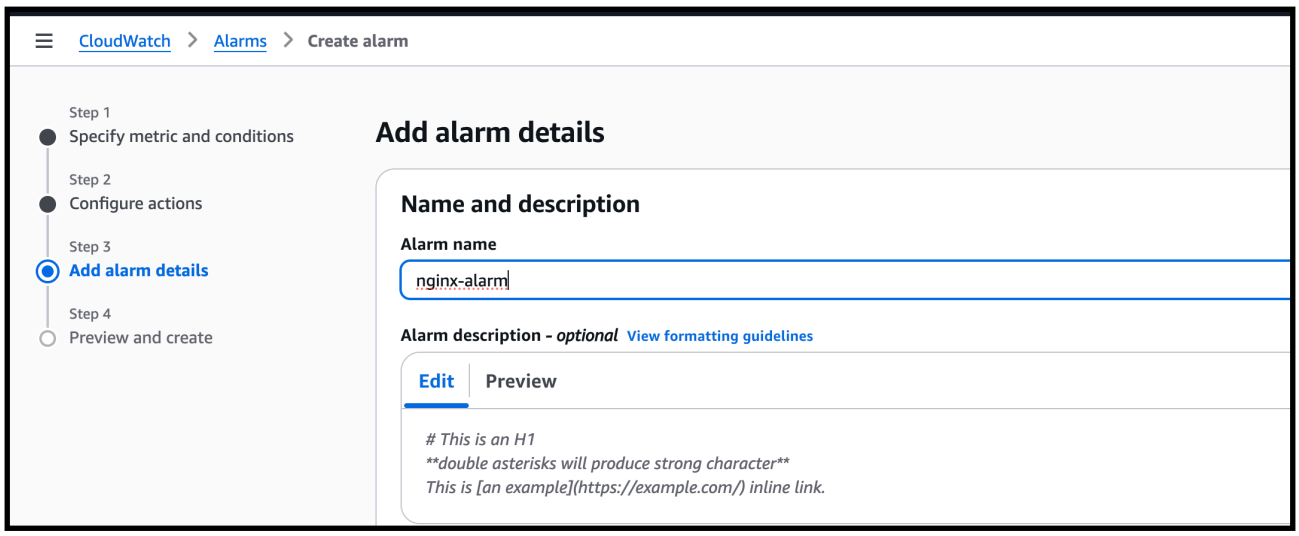
—-> now create the alarm for the nginx



—-> give same rules for nginx also if dashboard of the nginx is 0 trigger the alarm

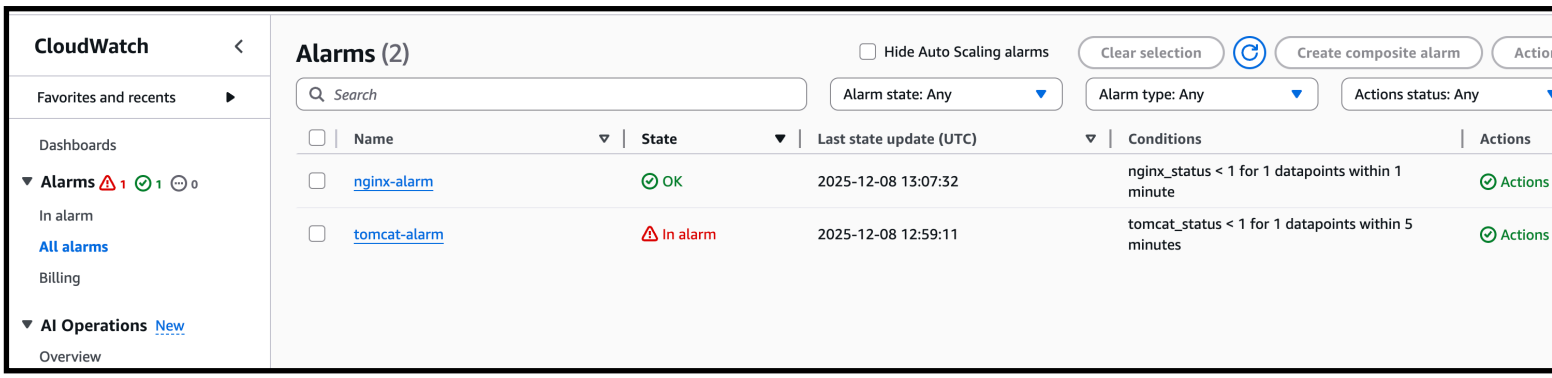


—> select the sns for email notification process

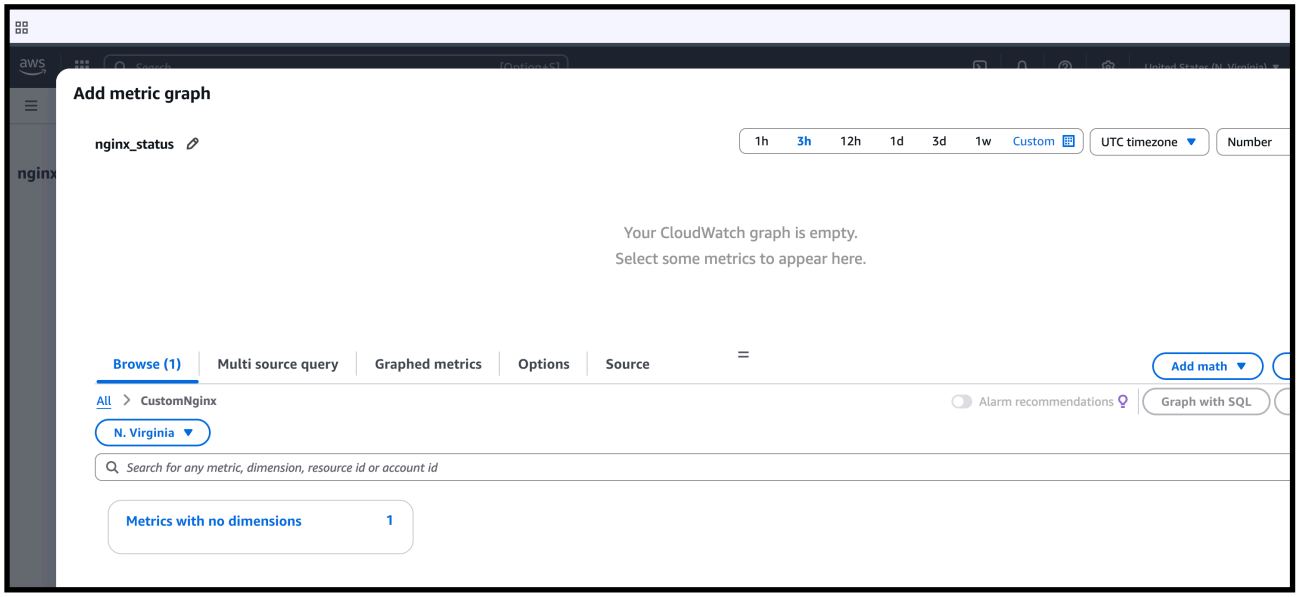


—> give the alarm name to nginx

—-> after alarm create we have set the alarm



—->to check aur nginx dashboard
—> selected the dashboard



—-> and select the matric it will show the dash board value as 1 because the nginx is running

—> nginx is running if the nginx is stoped the status will be 0
—> and we will get the alarm notification to the email same as tomcat from task 4